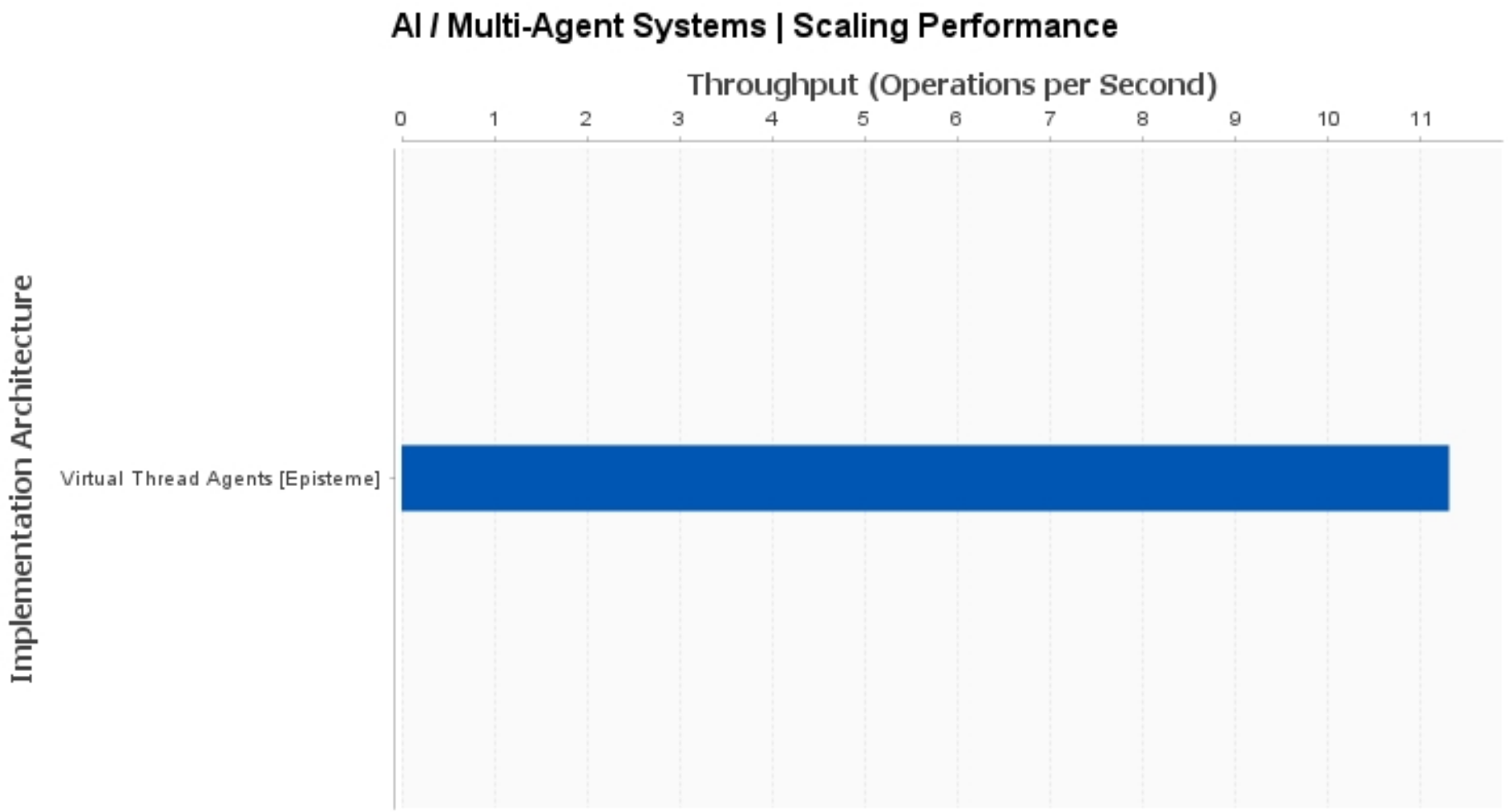
Episteme Benchmark Executive Summary

Precision Performance Analytics | Generated: 2026-03-04 00:37:51

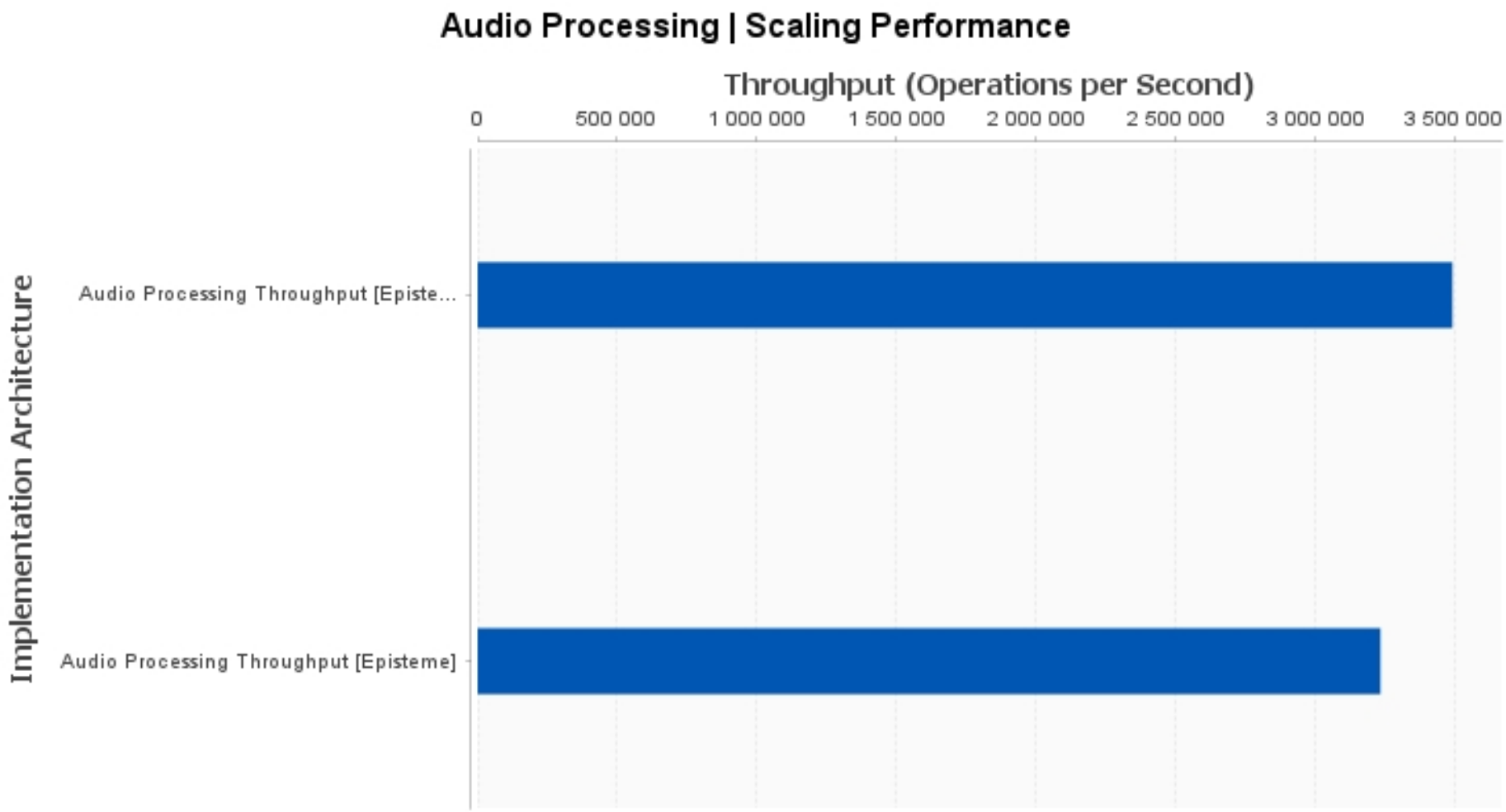
Operating System	Windows 11 (amd64)
Java Runtime	25 (Oracle Corporation)
Logical Processors	4

Domain Analysis: AI / Multi-Agent Systems

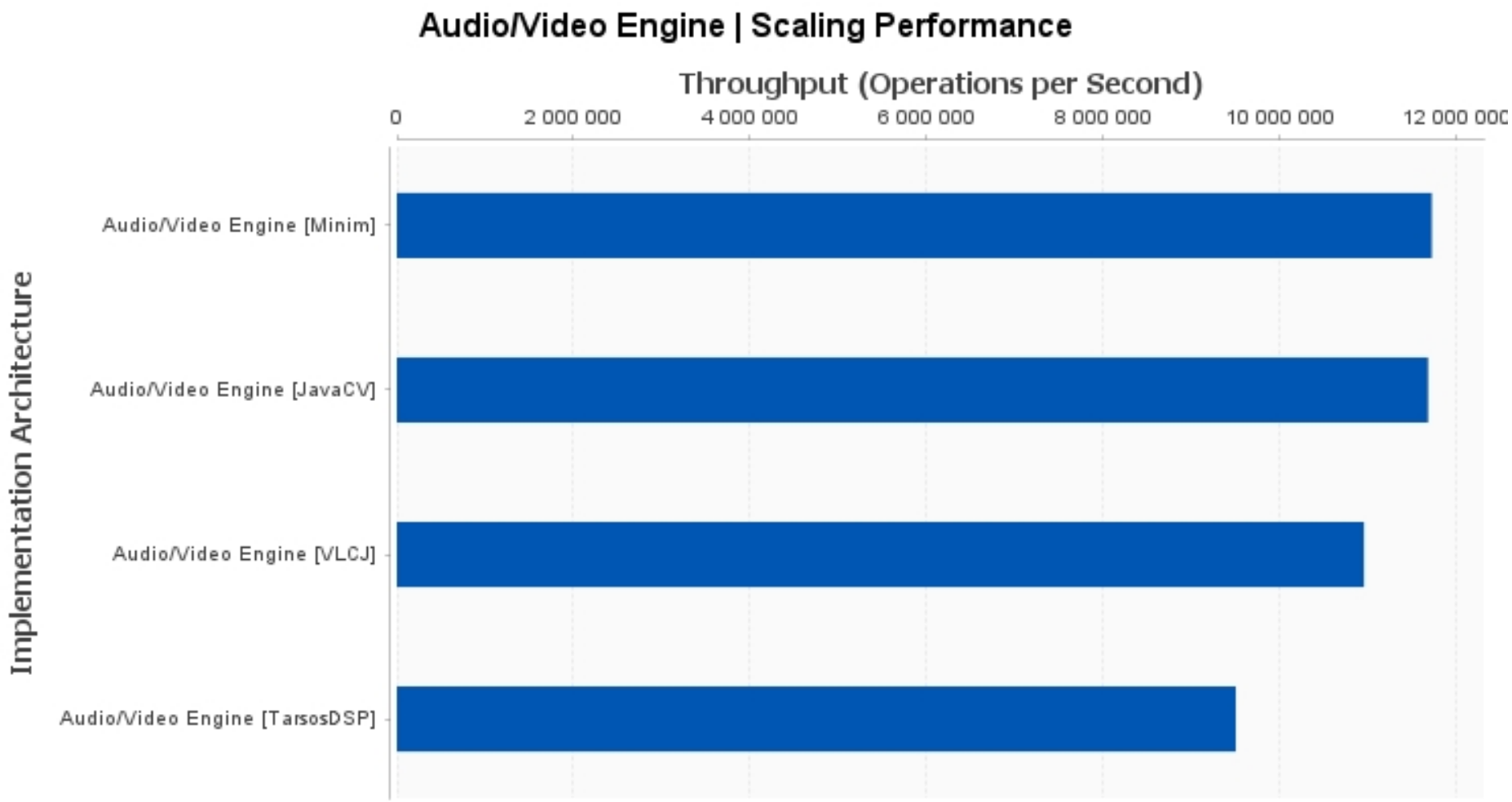
Domain Analysis: Audio Processing



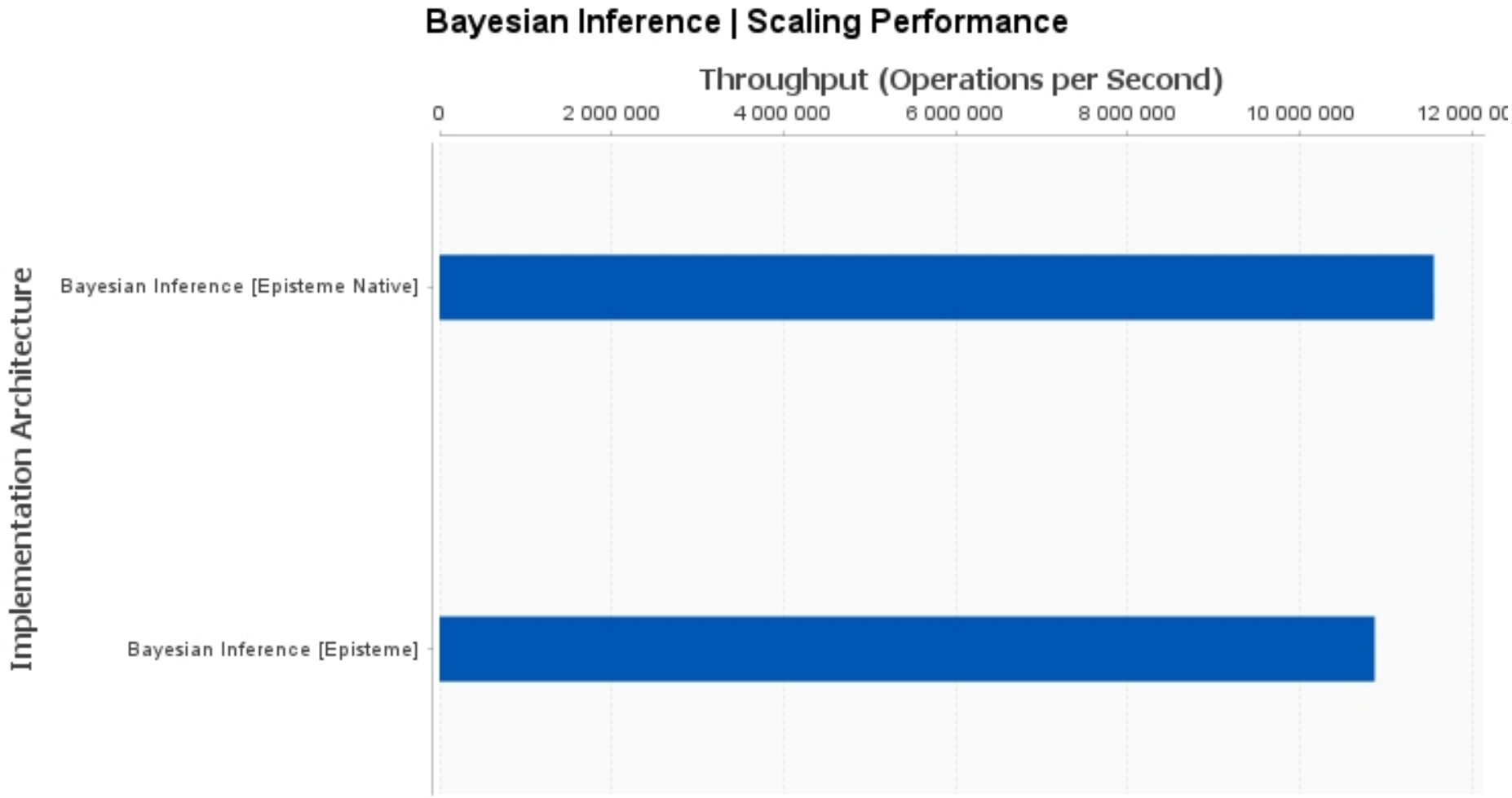
Domain Analysis: Audio/Video Engine



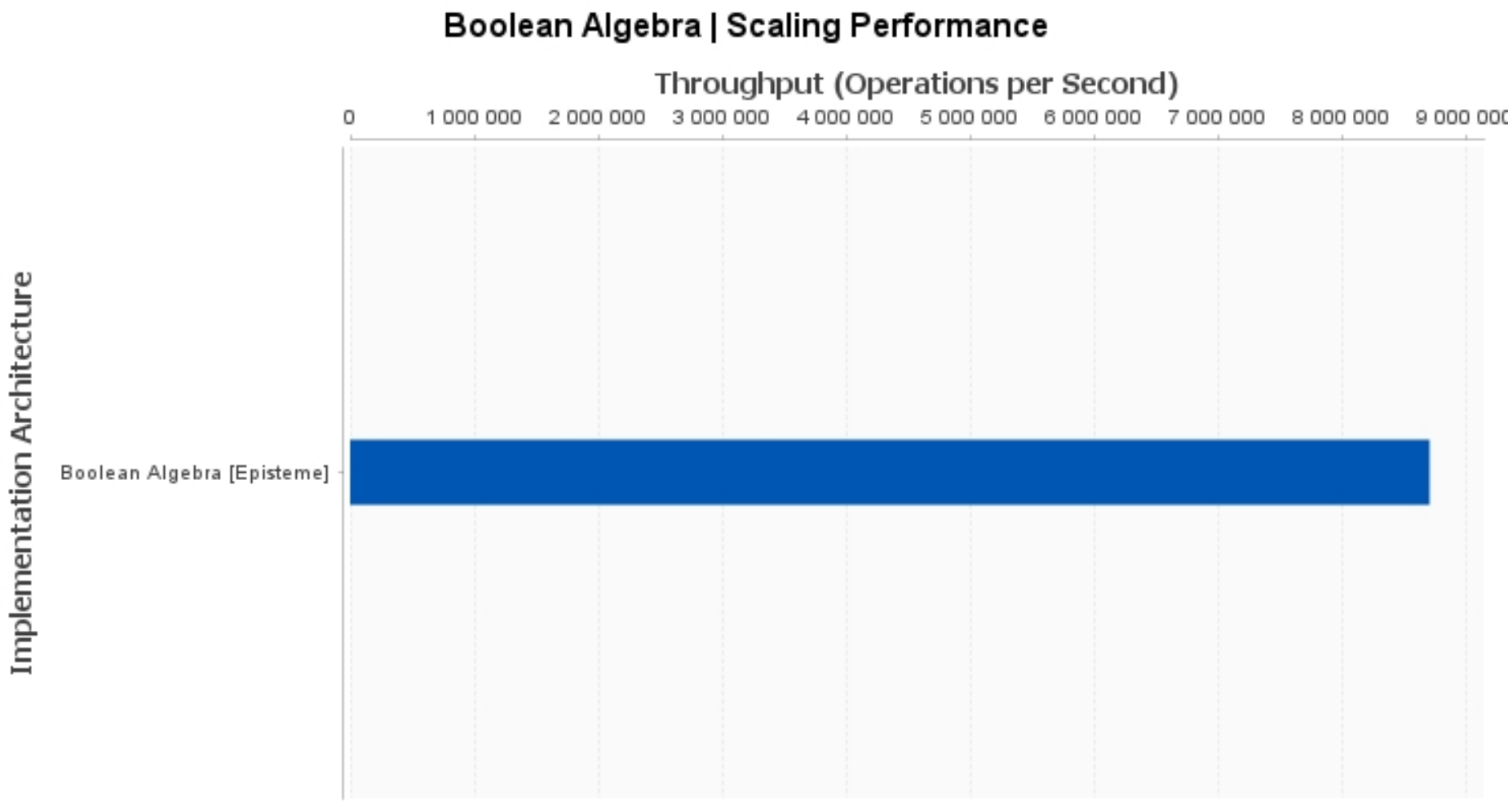
Domain Analysis: Bayesian Inference



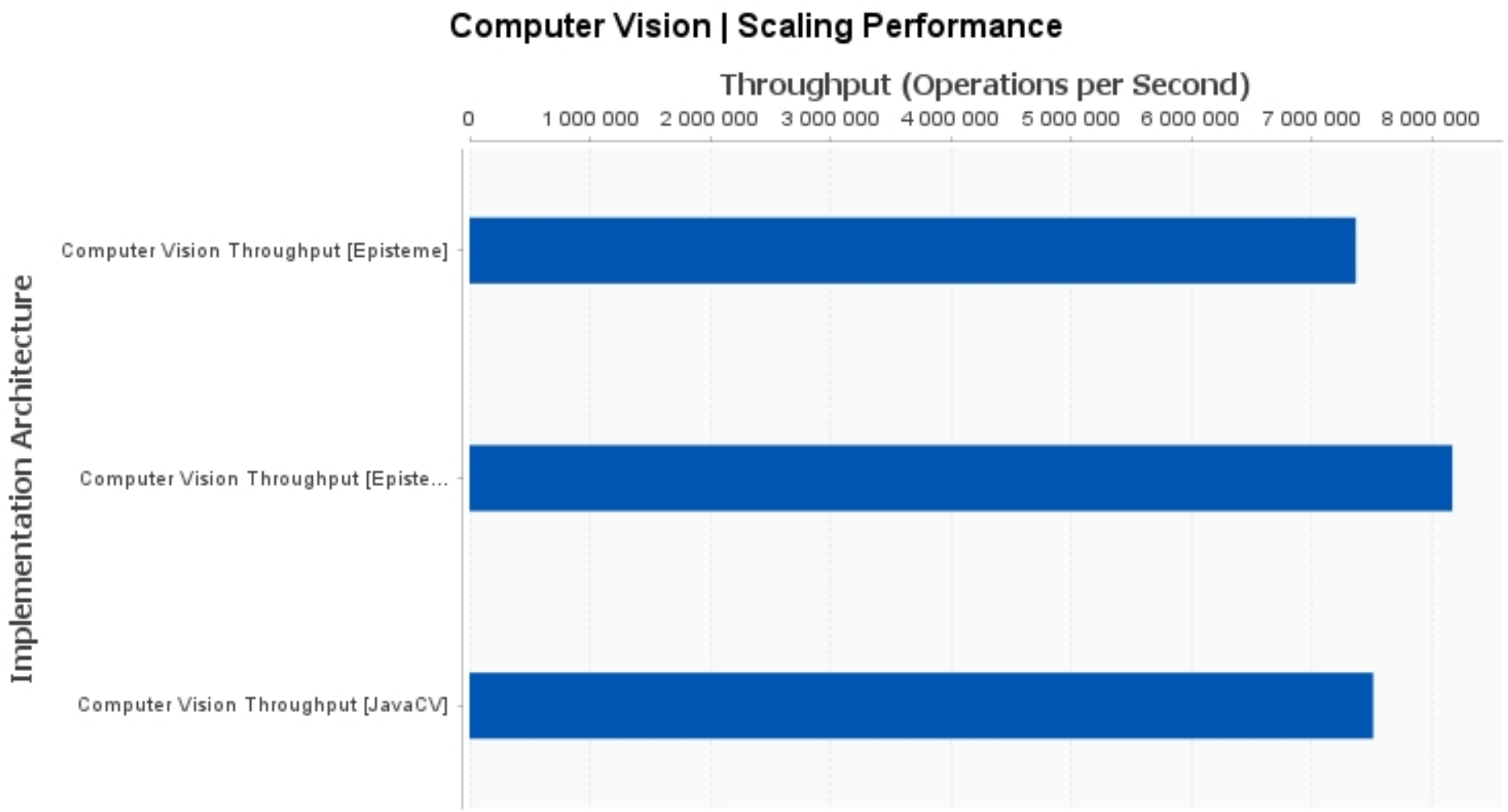
Domain Analysis: Boolean Algebra



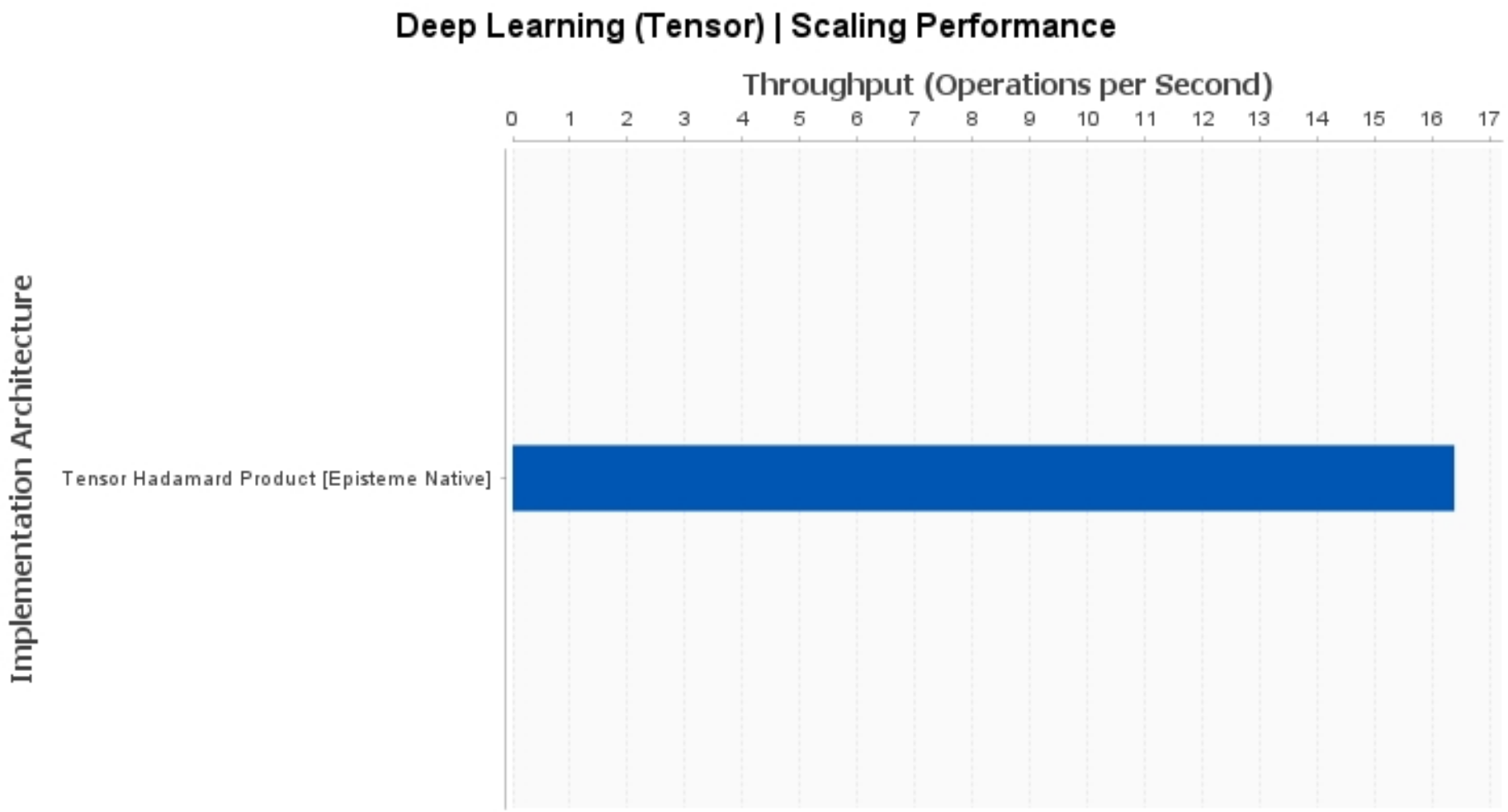
Domain Analysis: Computer Vision



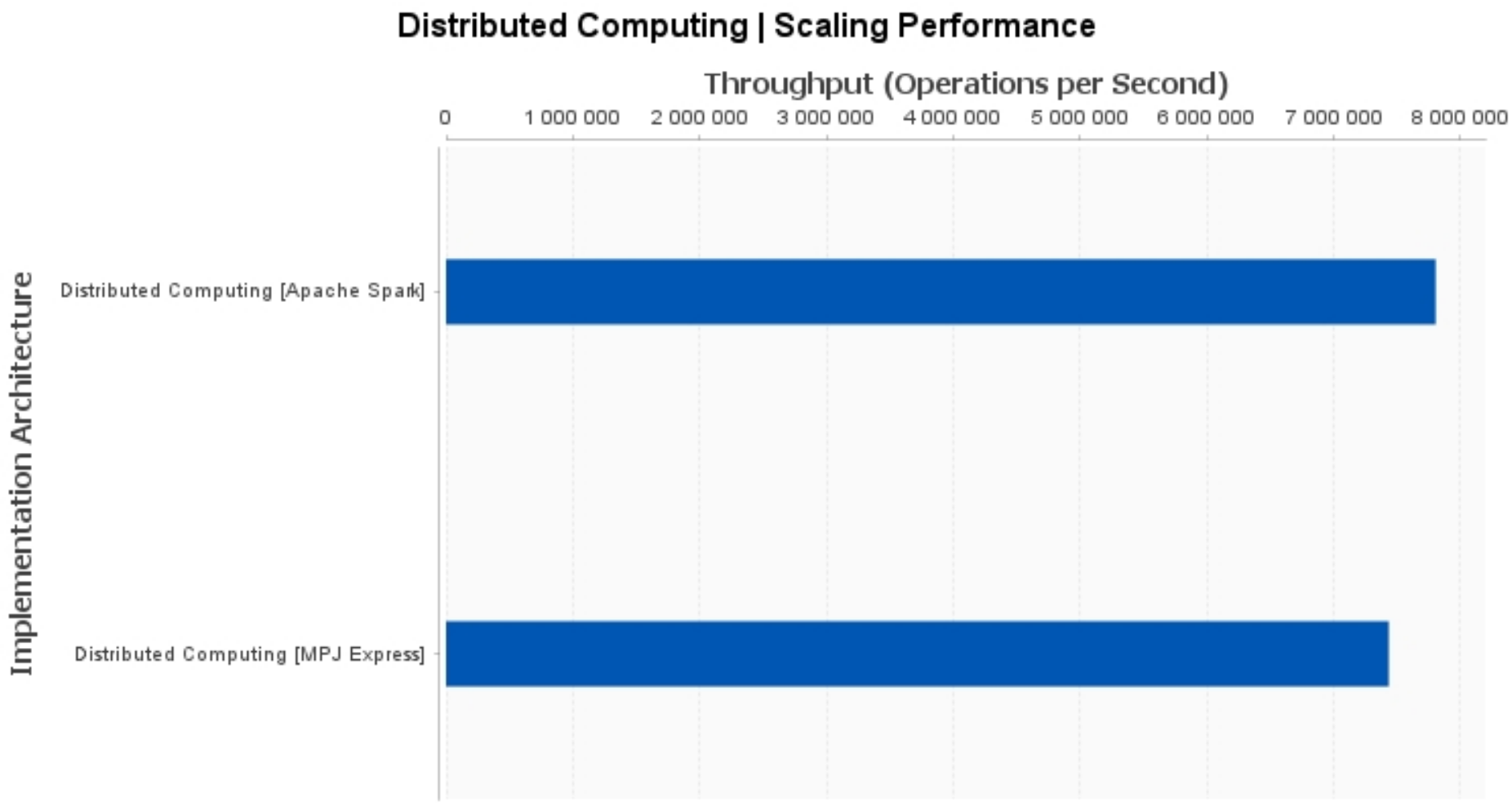
Domain Analysis: Deep Learning (Tensor)



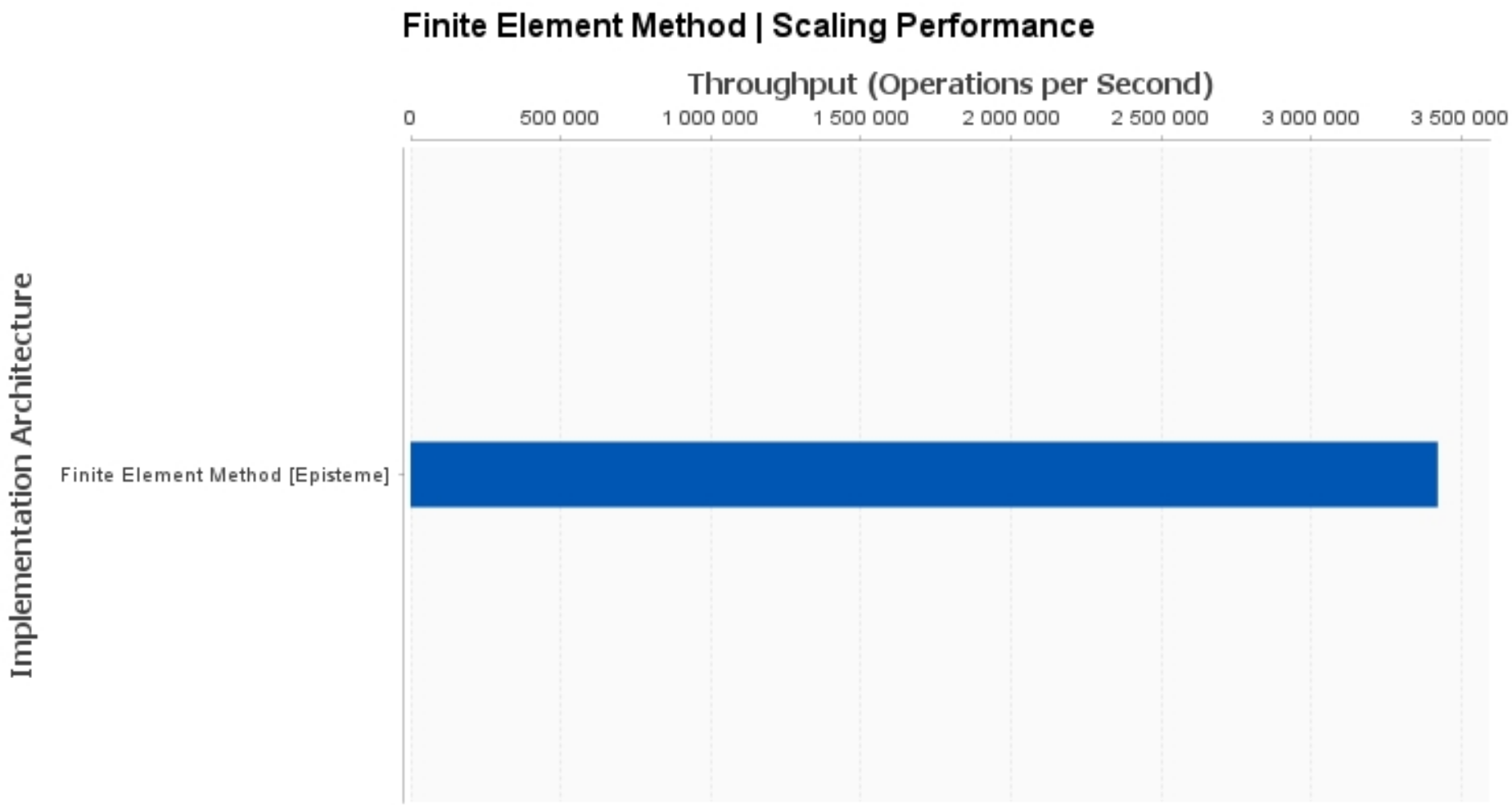
Domain Analysis: Distributed Computing



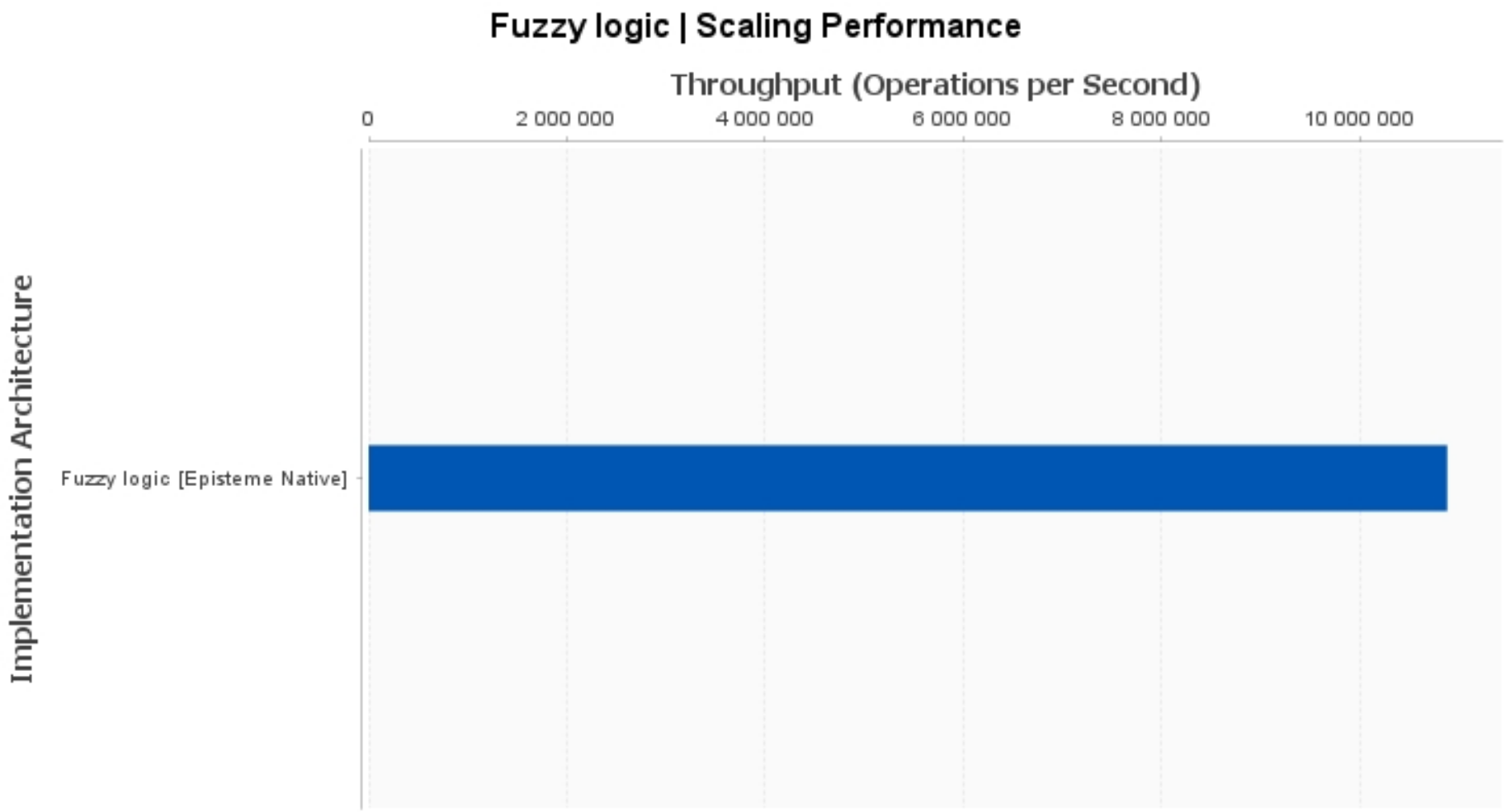
Domain Analysis: Finite Element Method



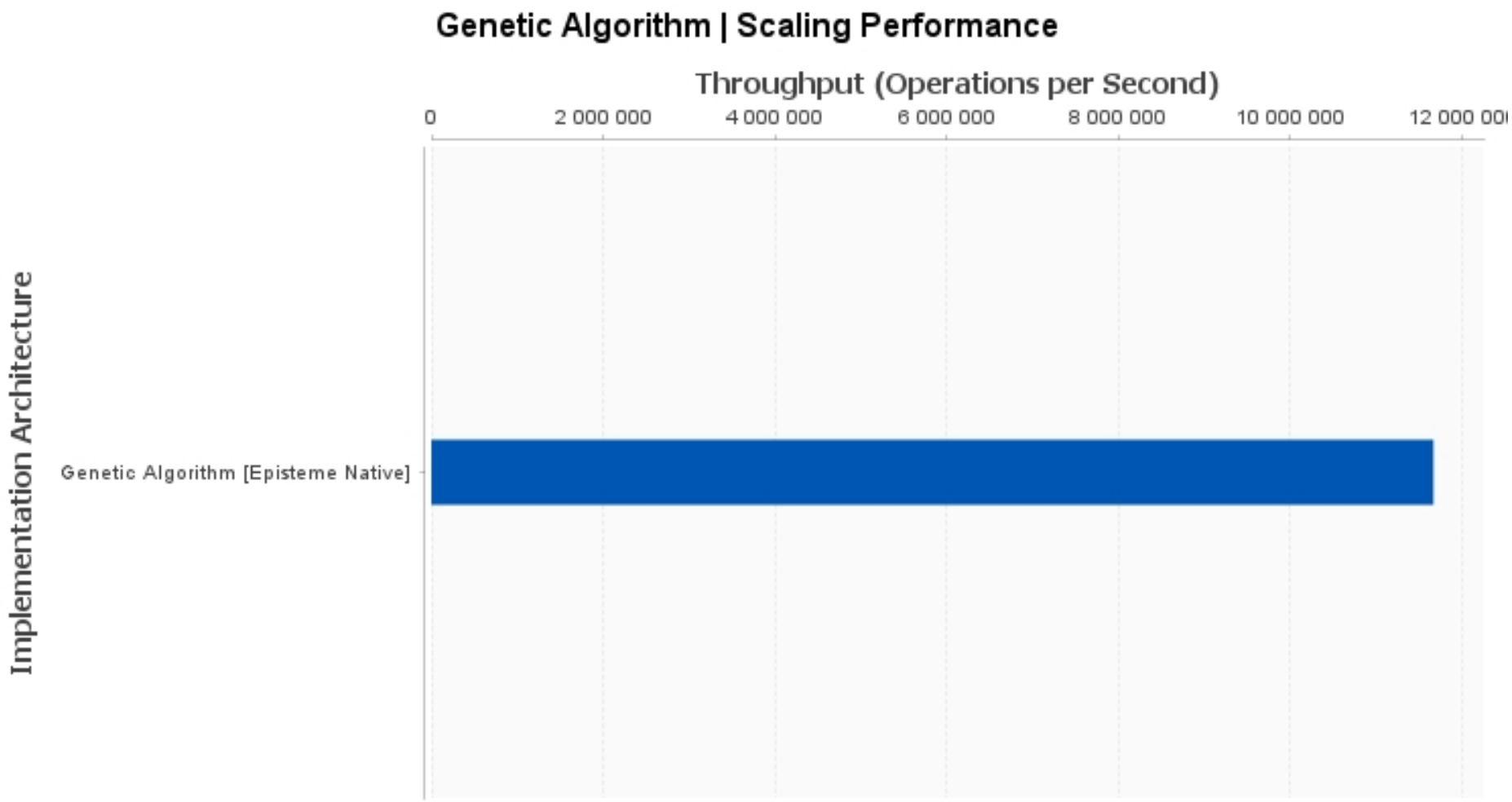
Domain Analysis: Fuzzy logic



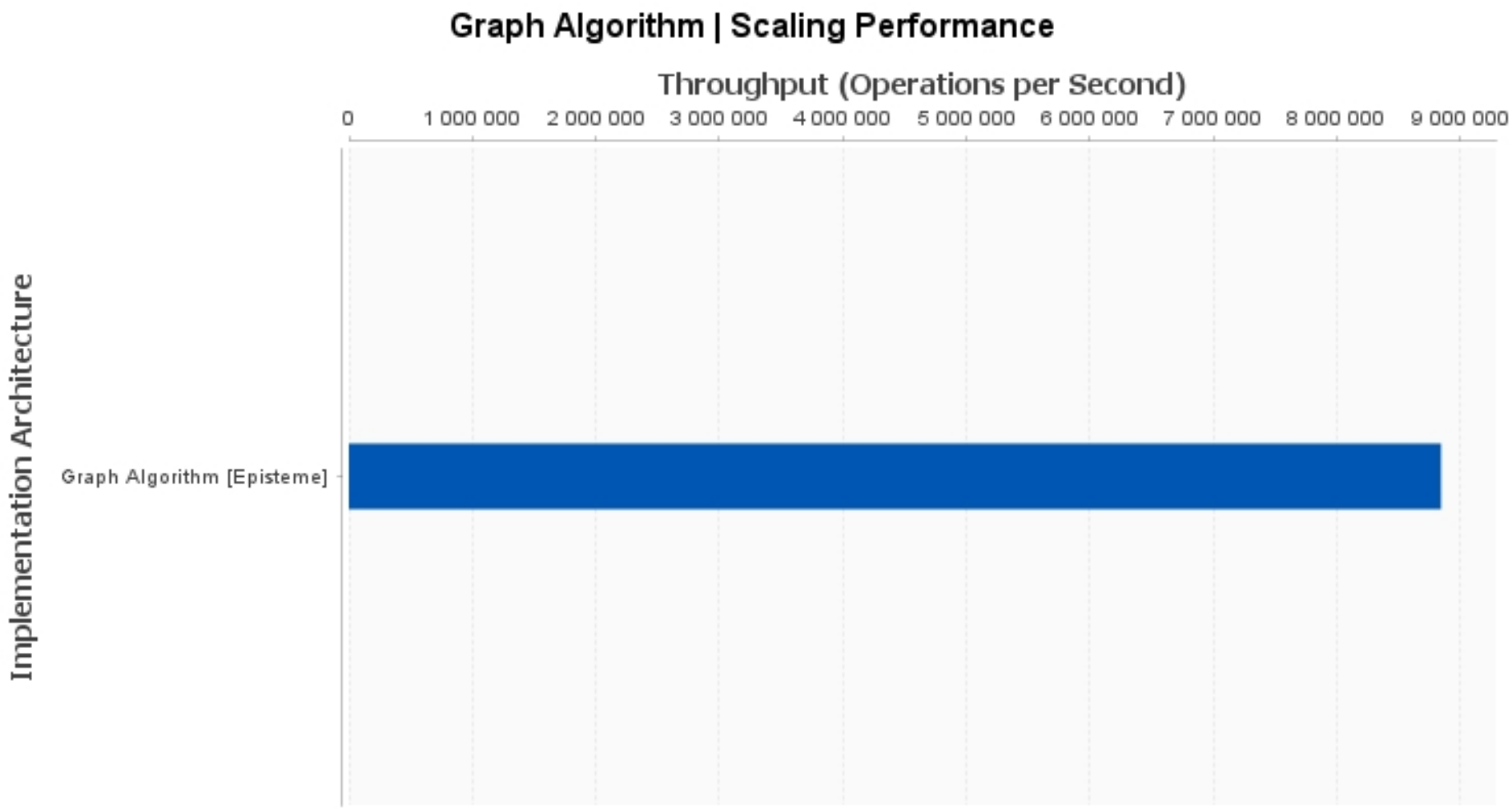
Domain Analysis: Genetic Algorithm



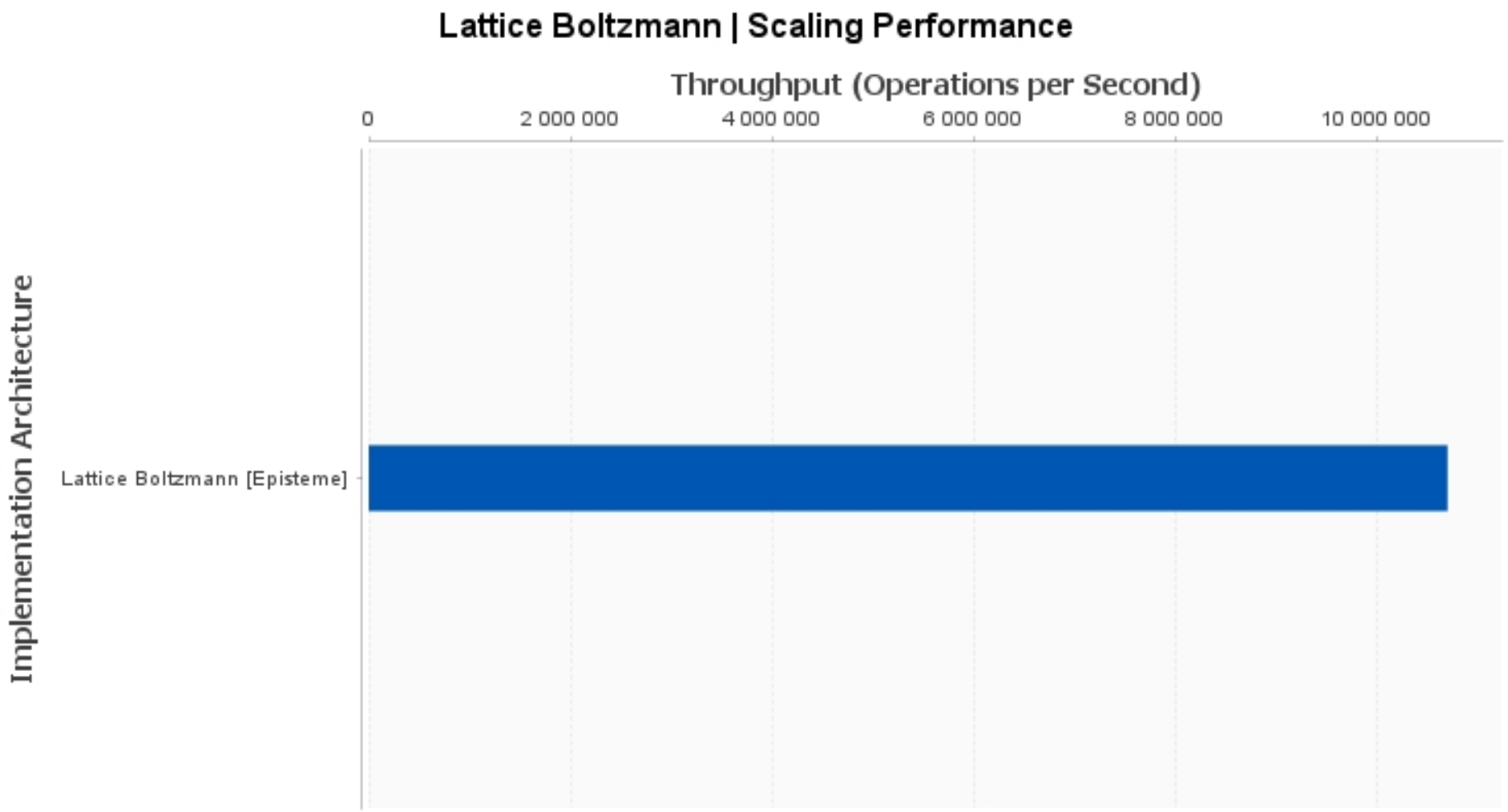
Domain Analysis: Graph Algorithm



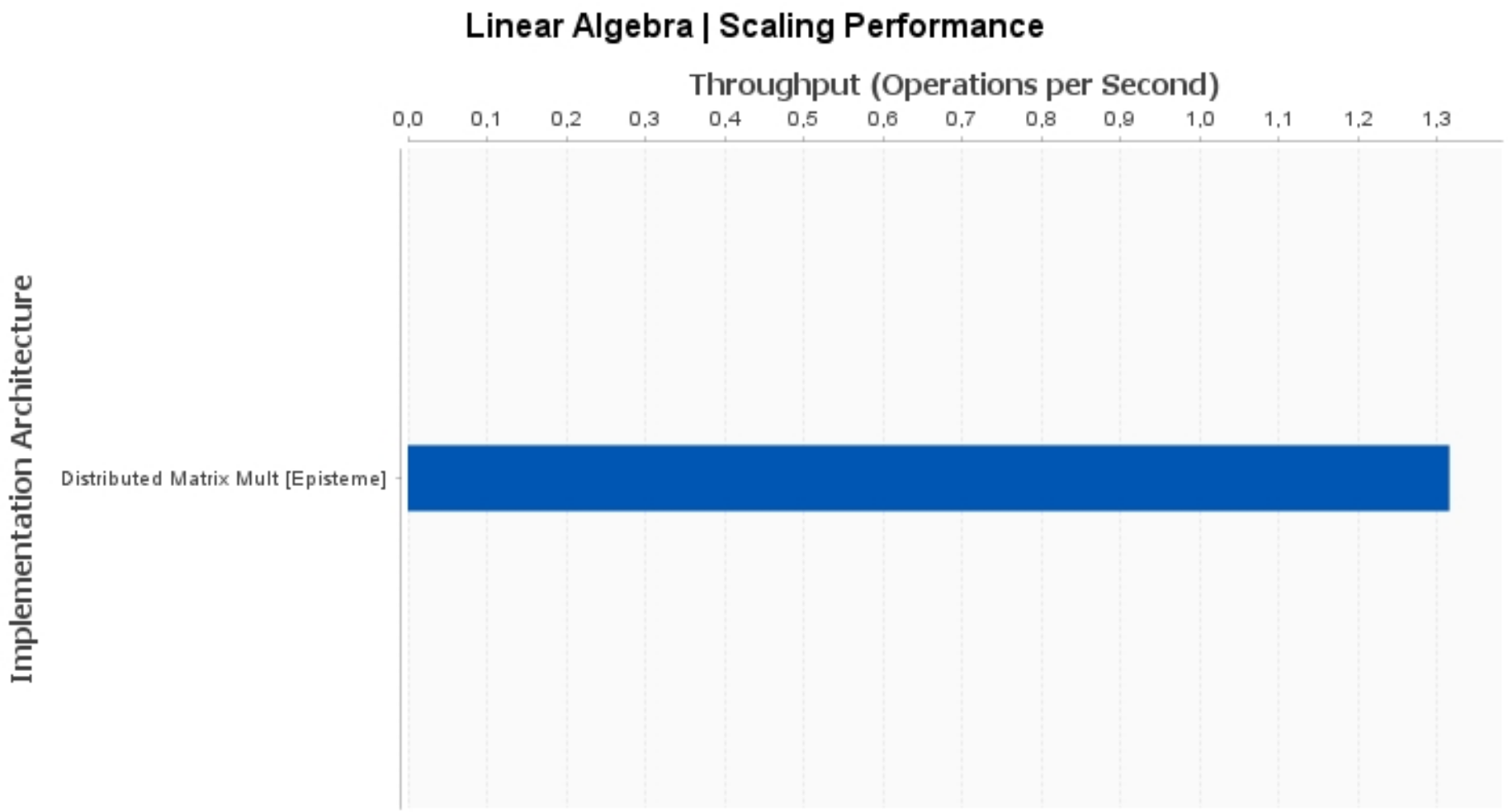
Domain Analysis: Lattice Boltzmann



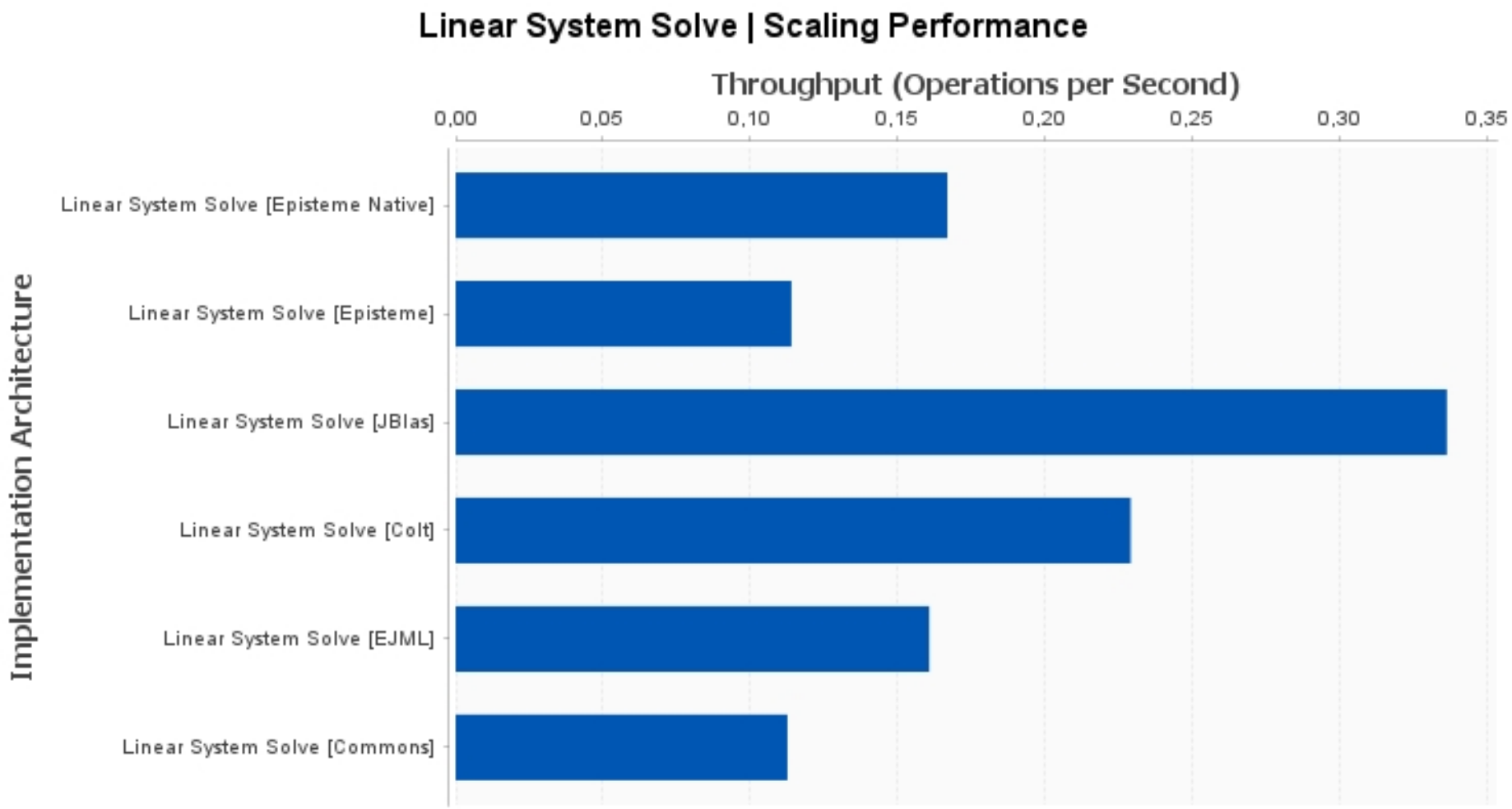
Domain Analysis: Linear Algebra



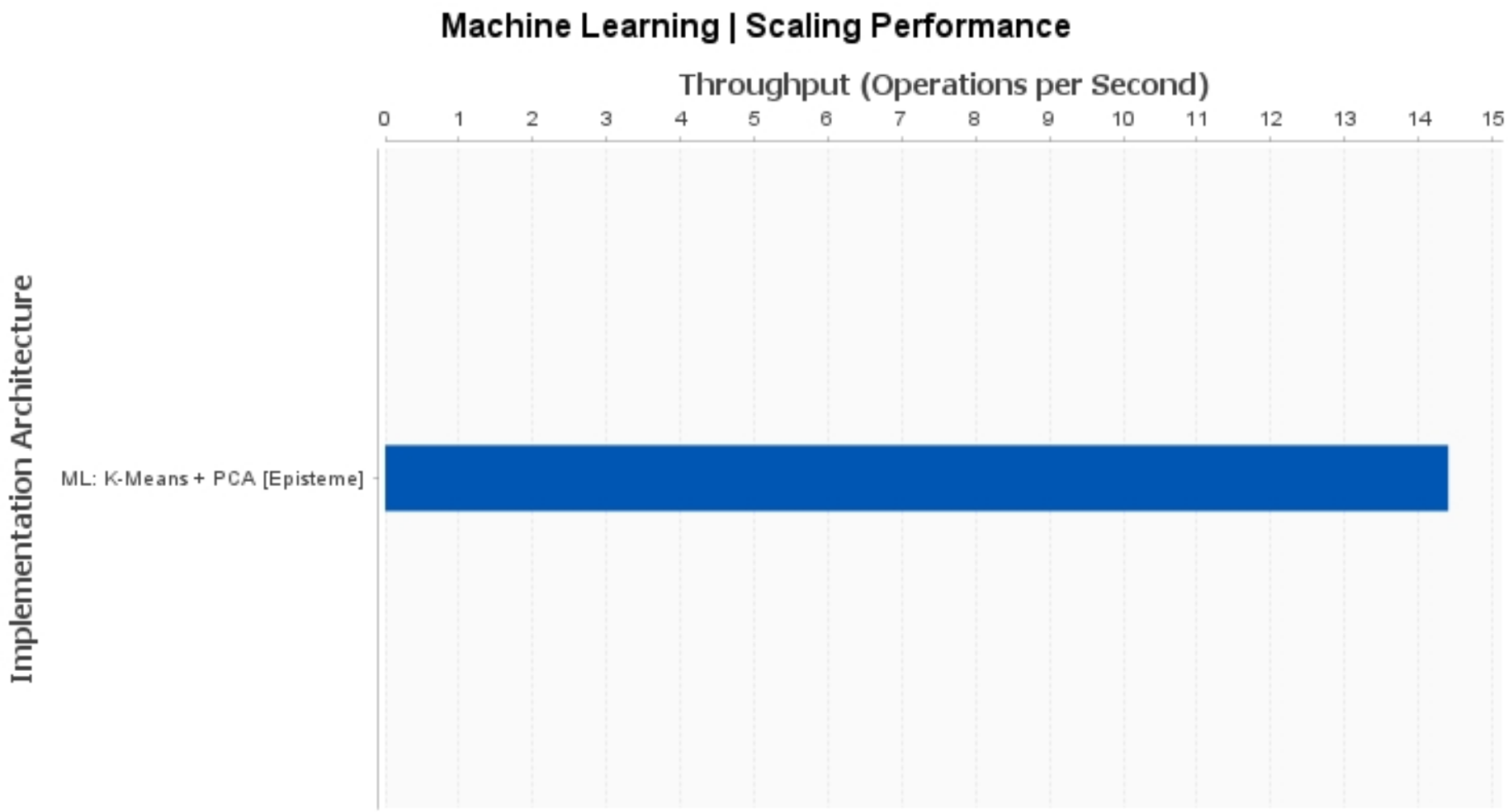
Domain Analysis: Linear System Solve



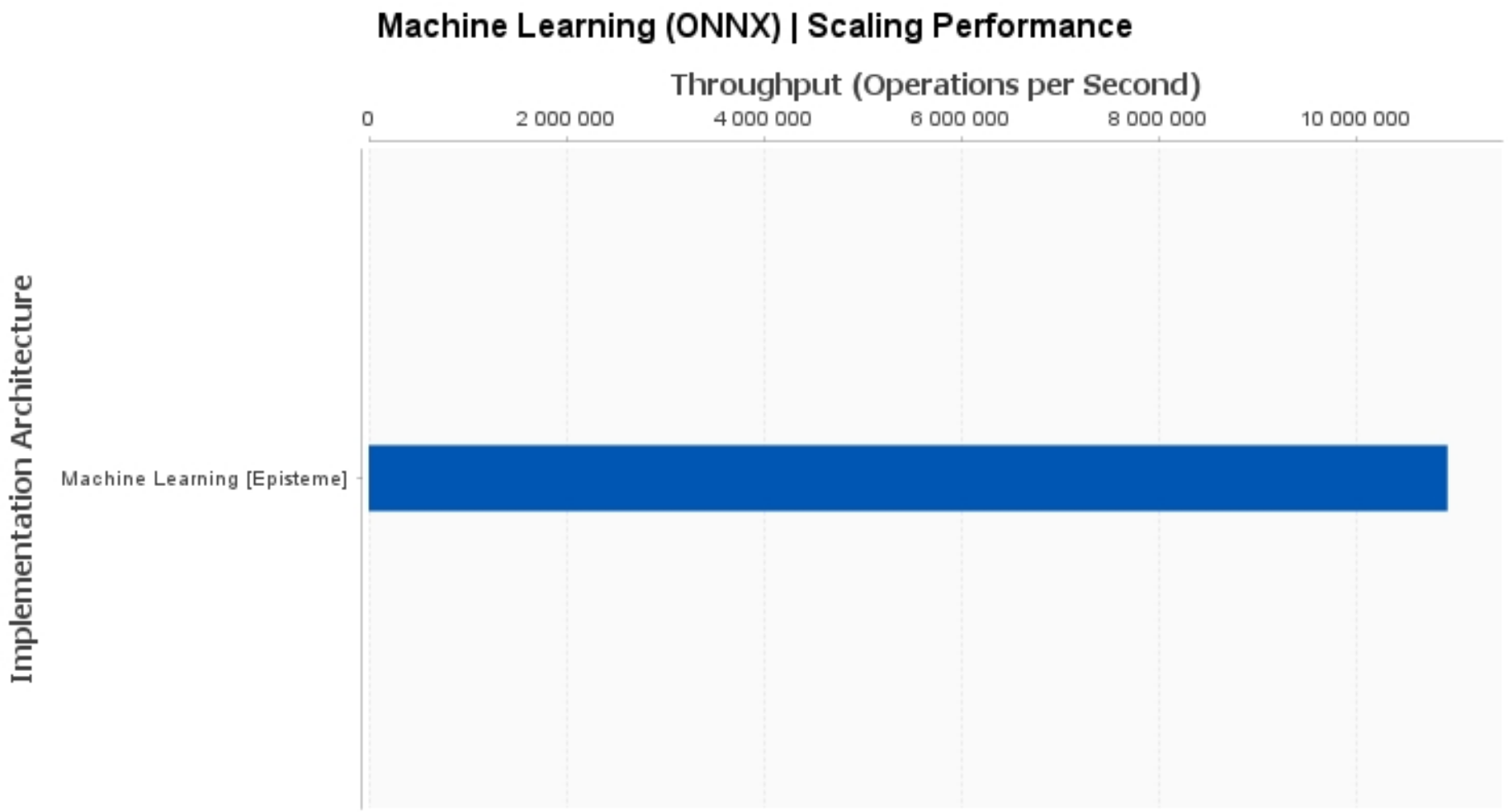
Domain Analysis: Machine Learning



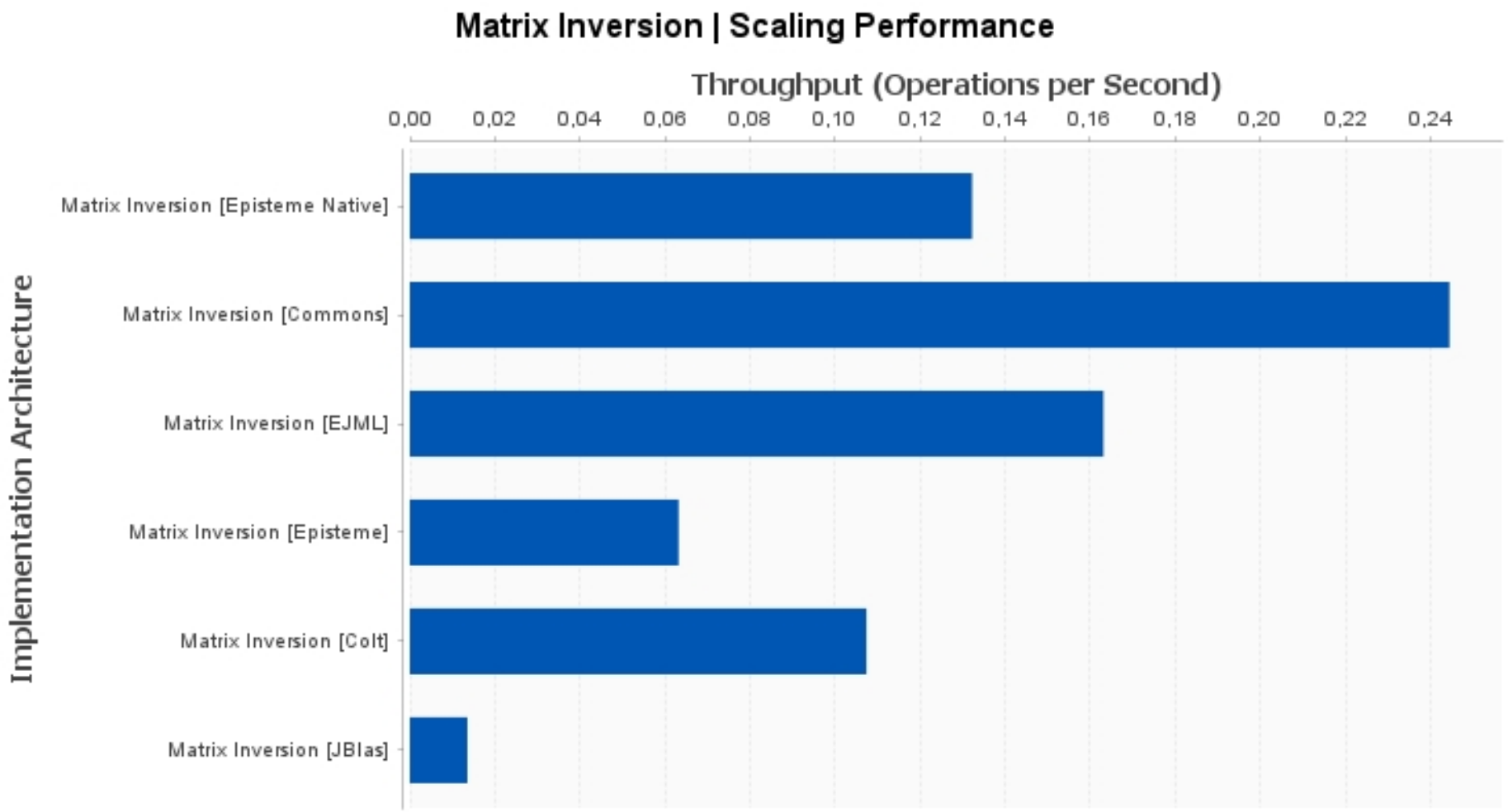
Domain Analysis: Machine Learning (ONNX)



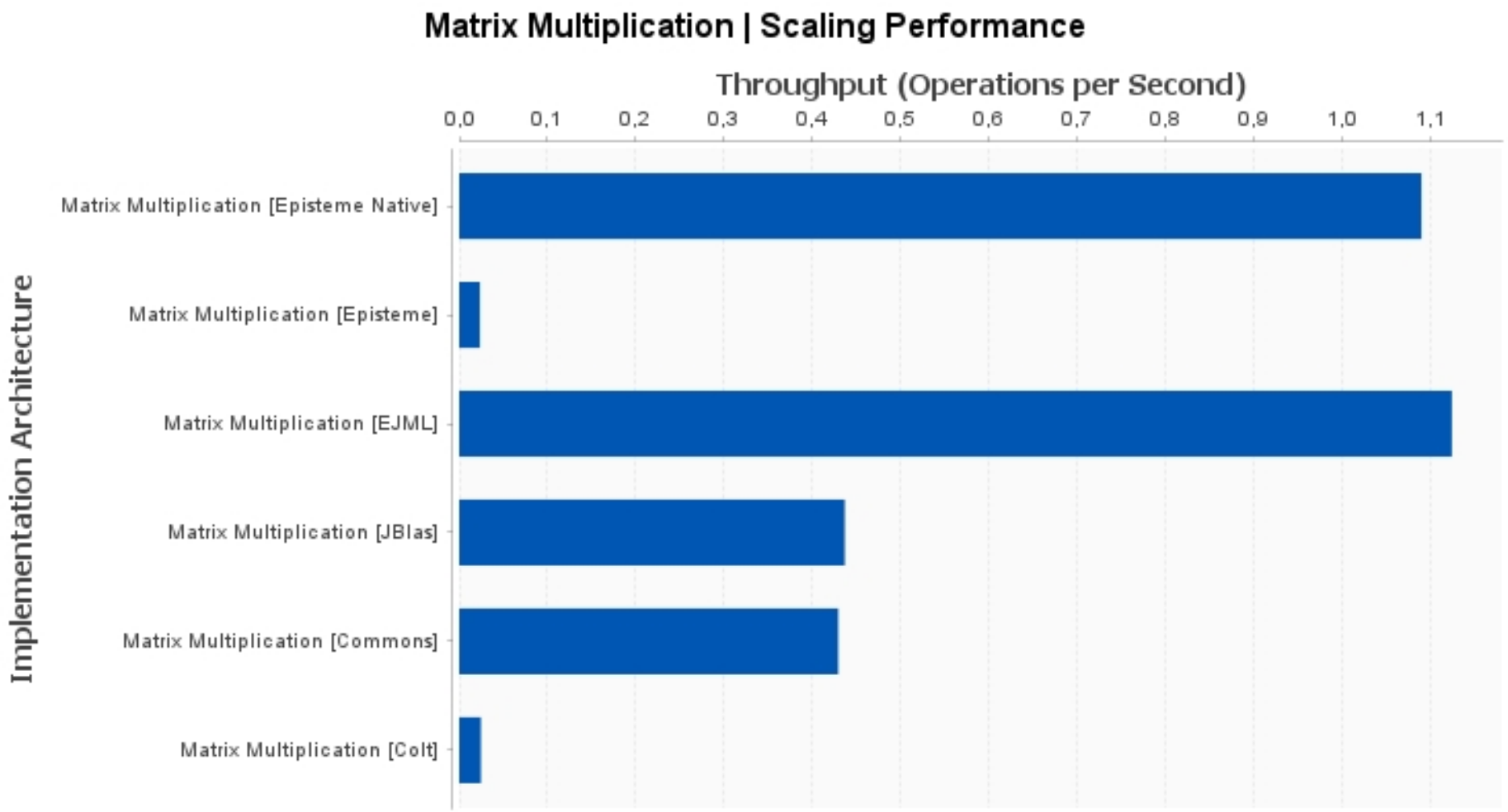
Domain Analysis: Matrix Inversion



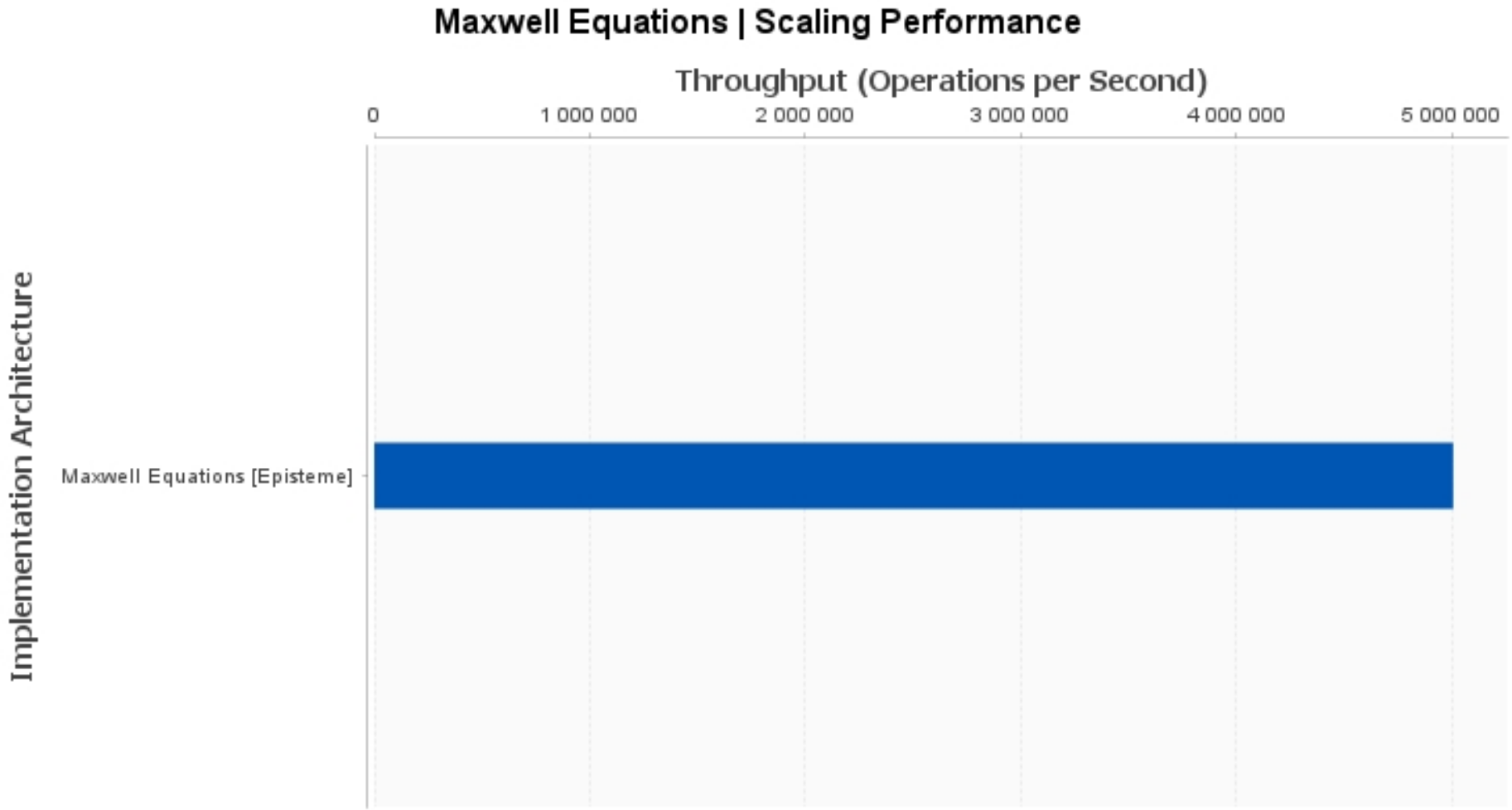
Domain Analysis: Matrix Multiplication



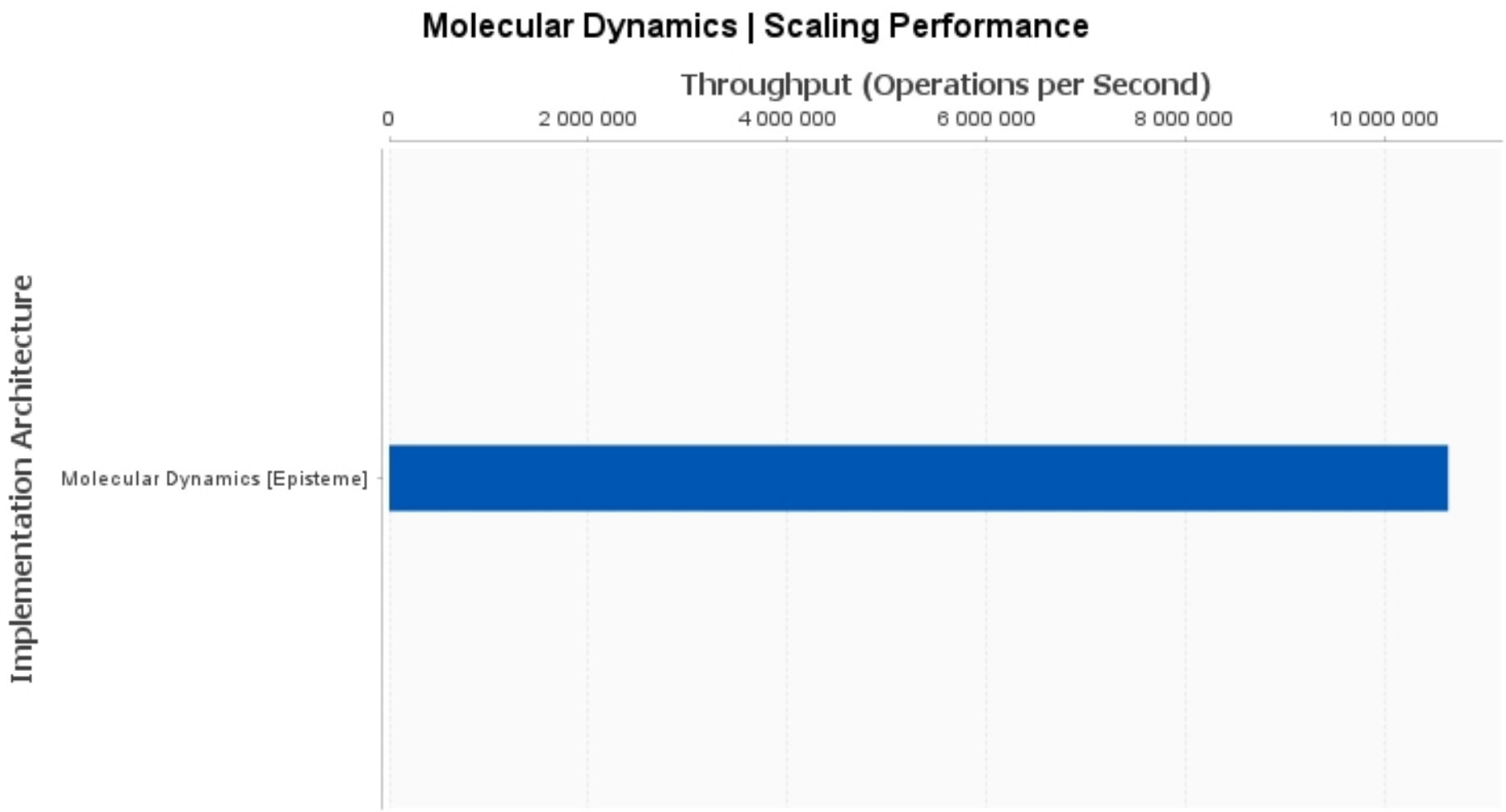
Domain Analysis: Maxwell Equations



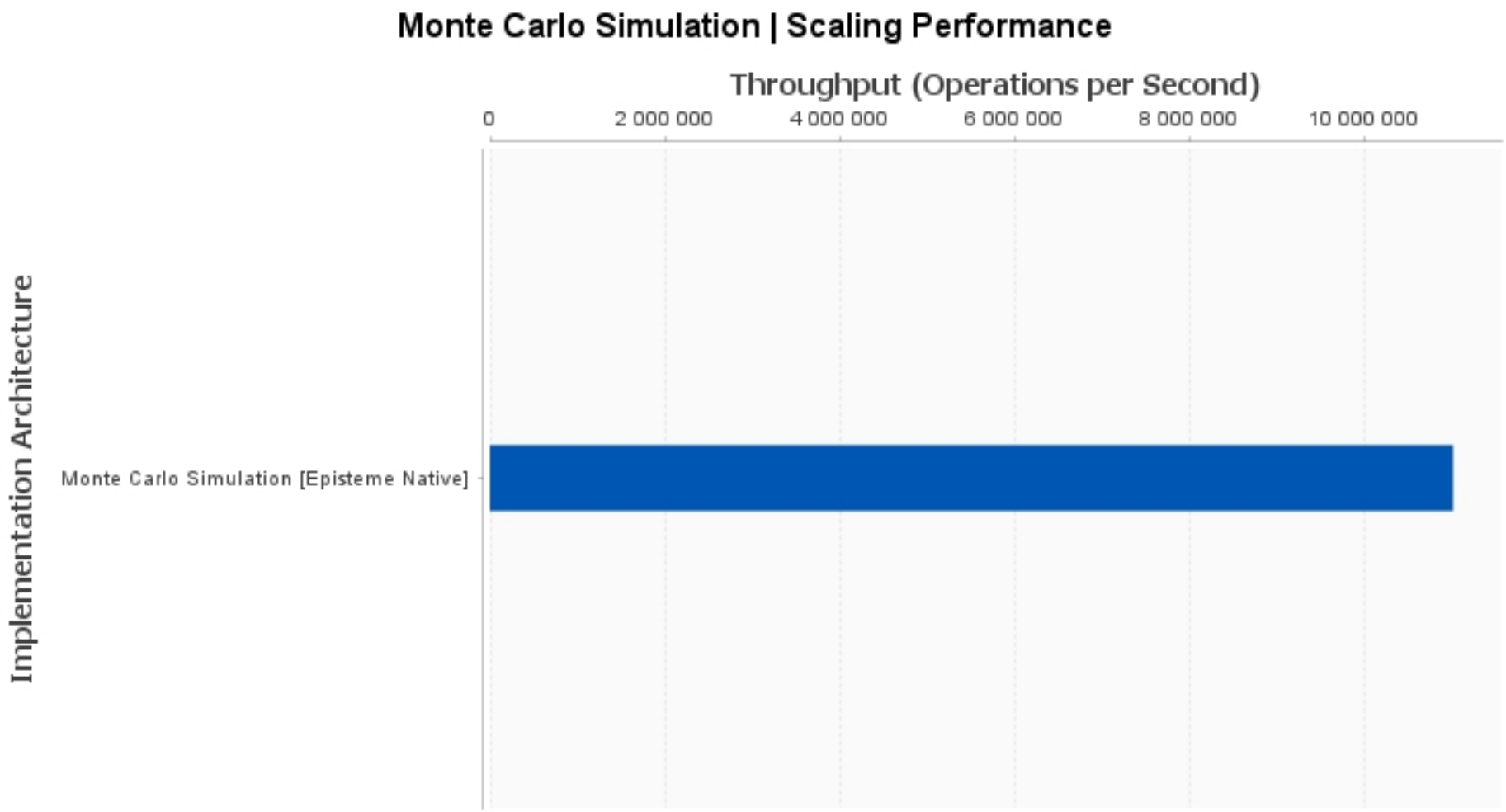
Domain Analysis: Molecular Dynamics



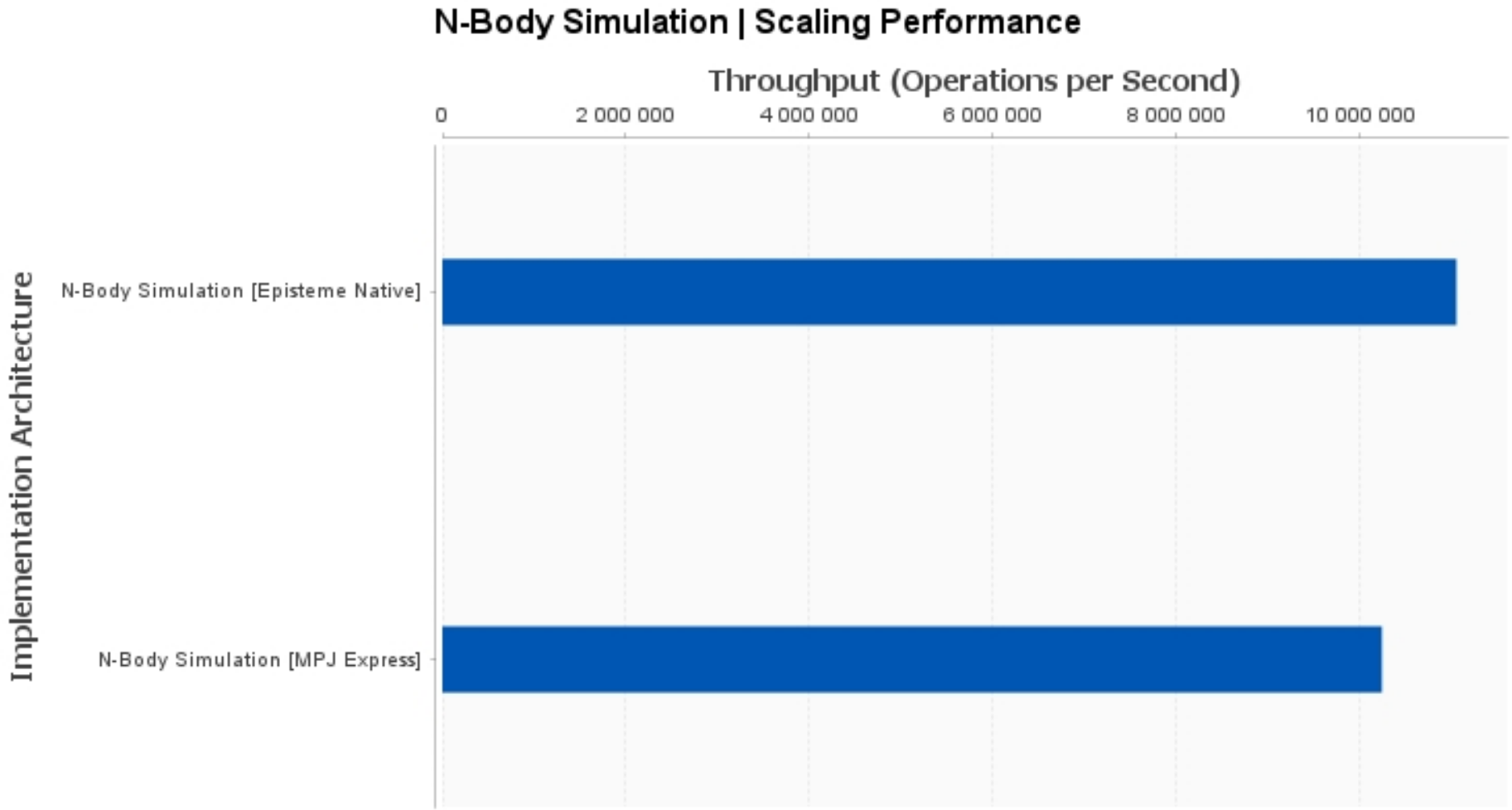
Domain Analysis: Monte Carlo Simulation



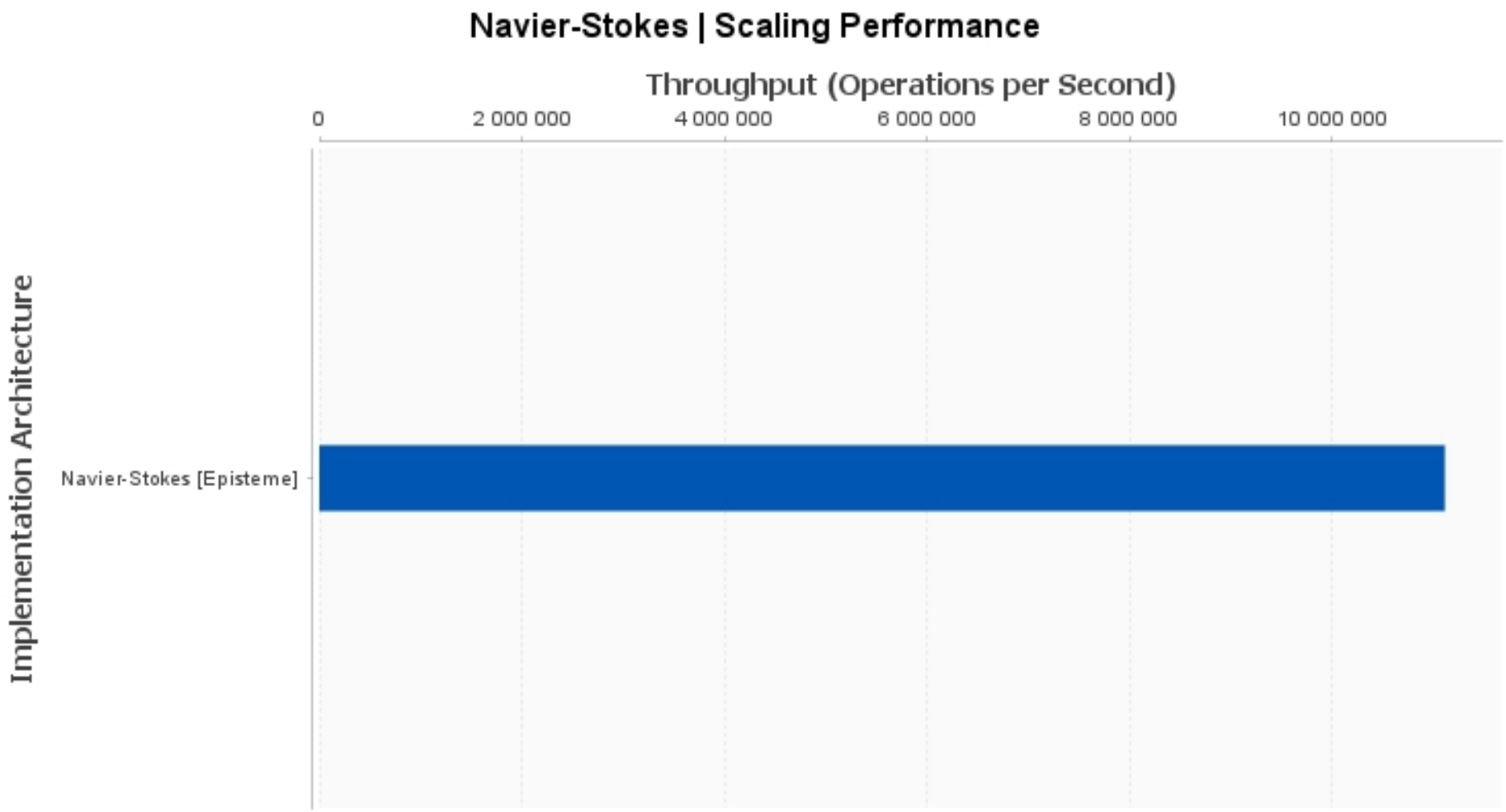
Domain Analysis: N-Body Simulation



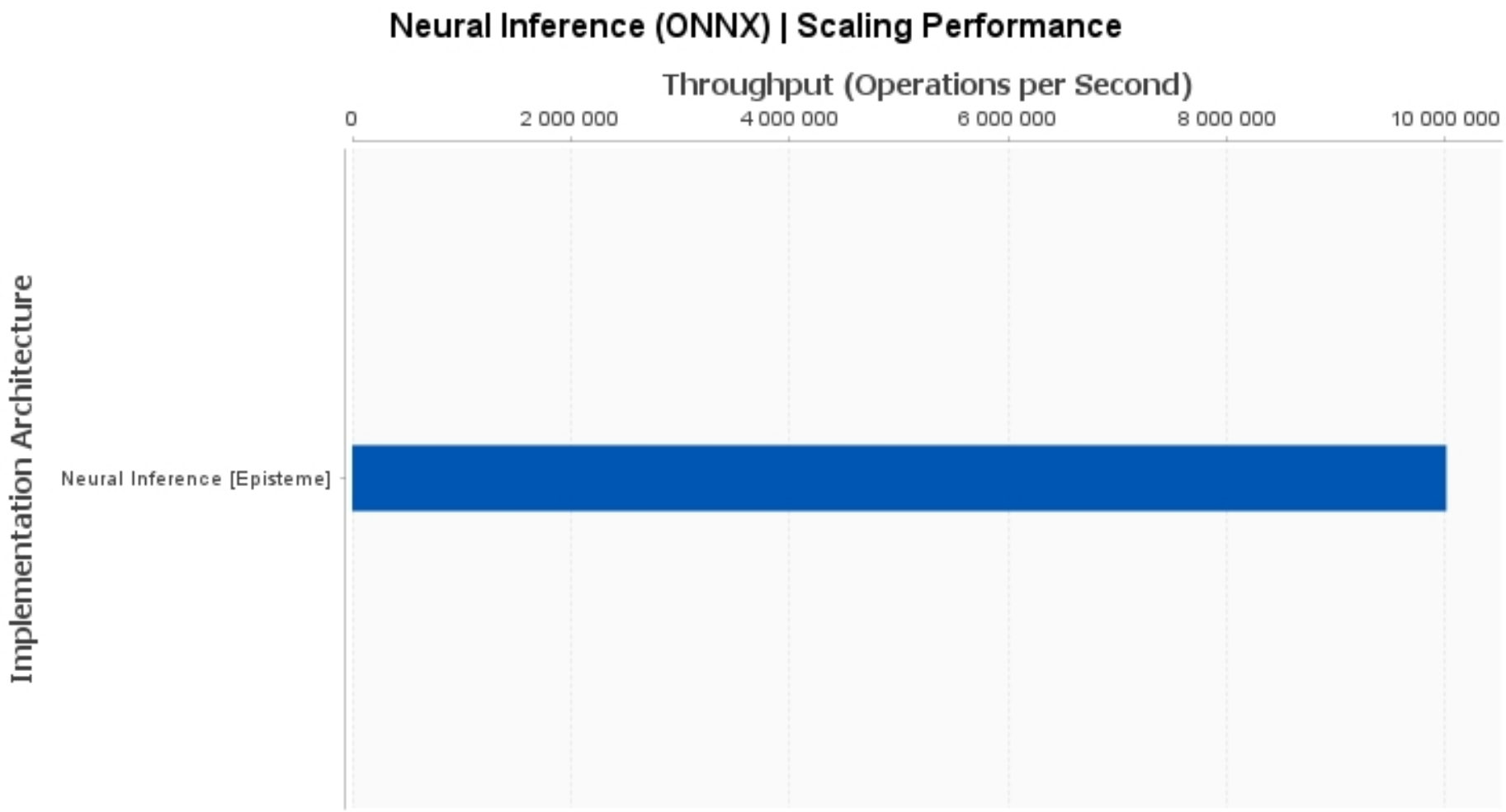
Domain Analysis: Navier-Stokes



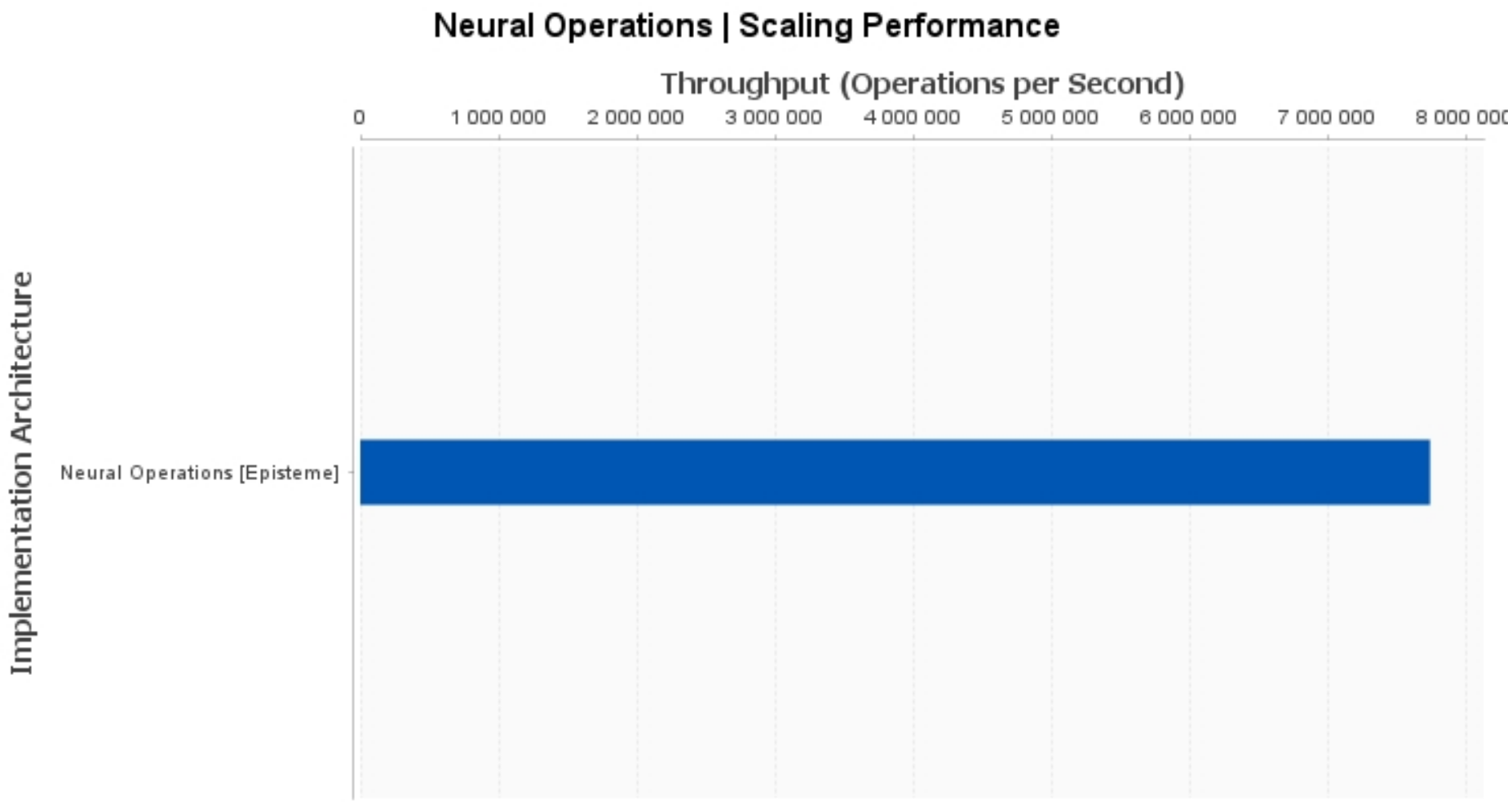
Domain Analysis: Neural Inference (ONNX)



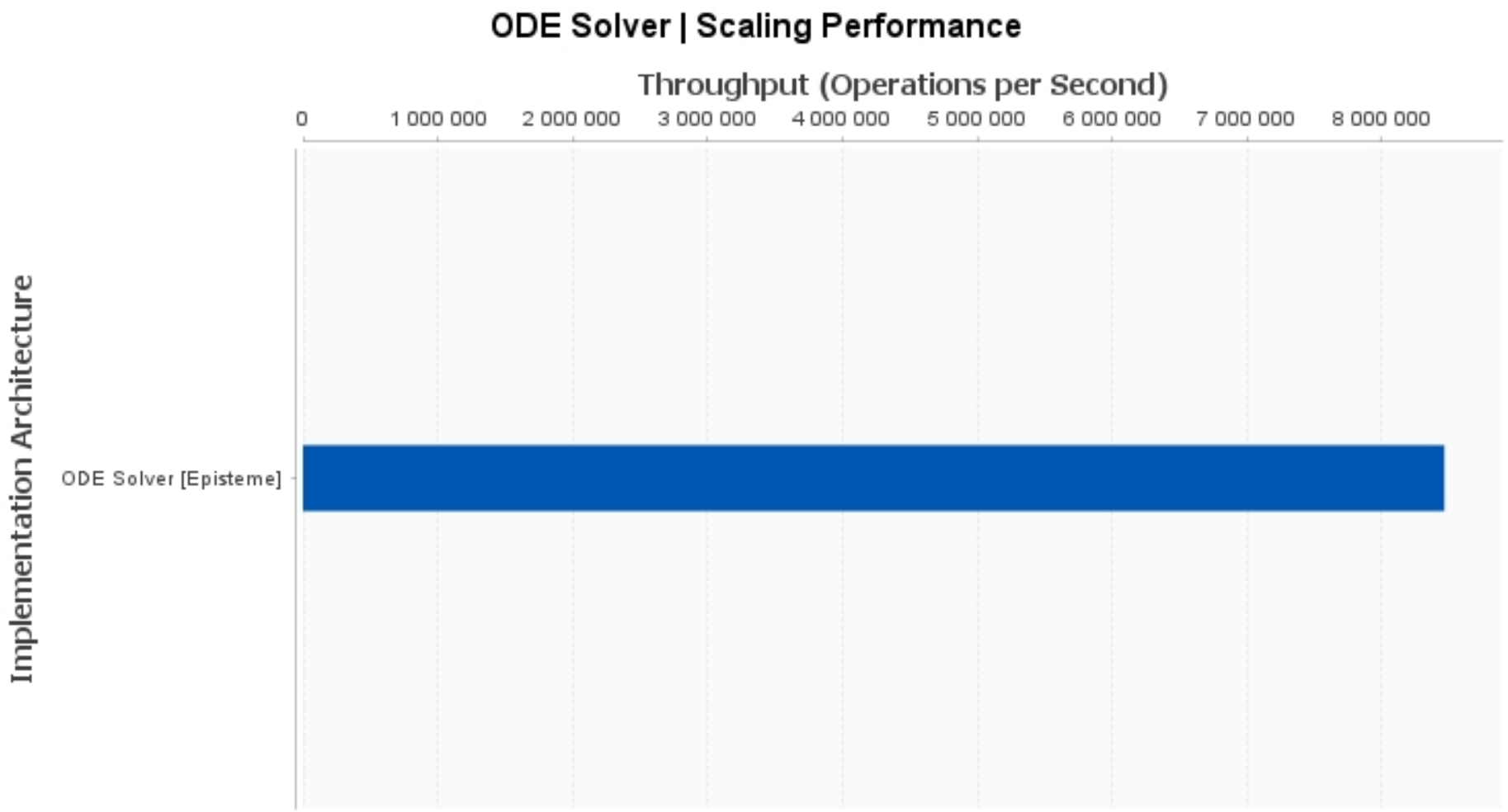
Domain Analysis: Neural Operations



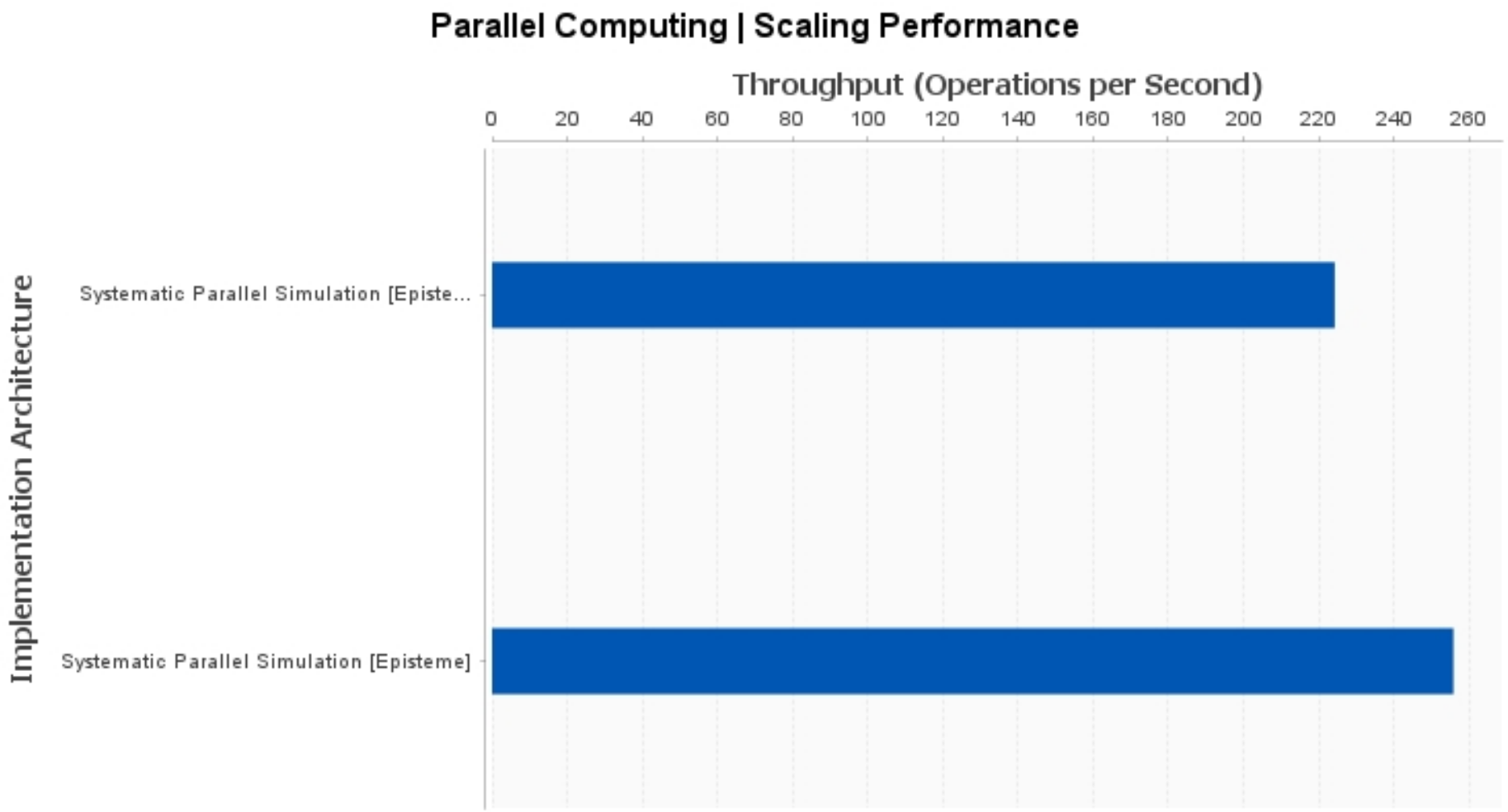
Domain Analysis: ODE Solver



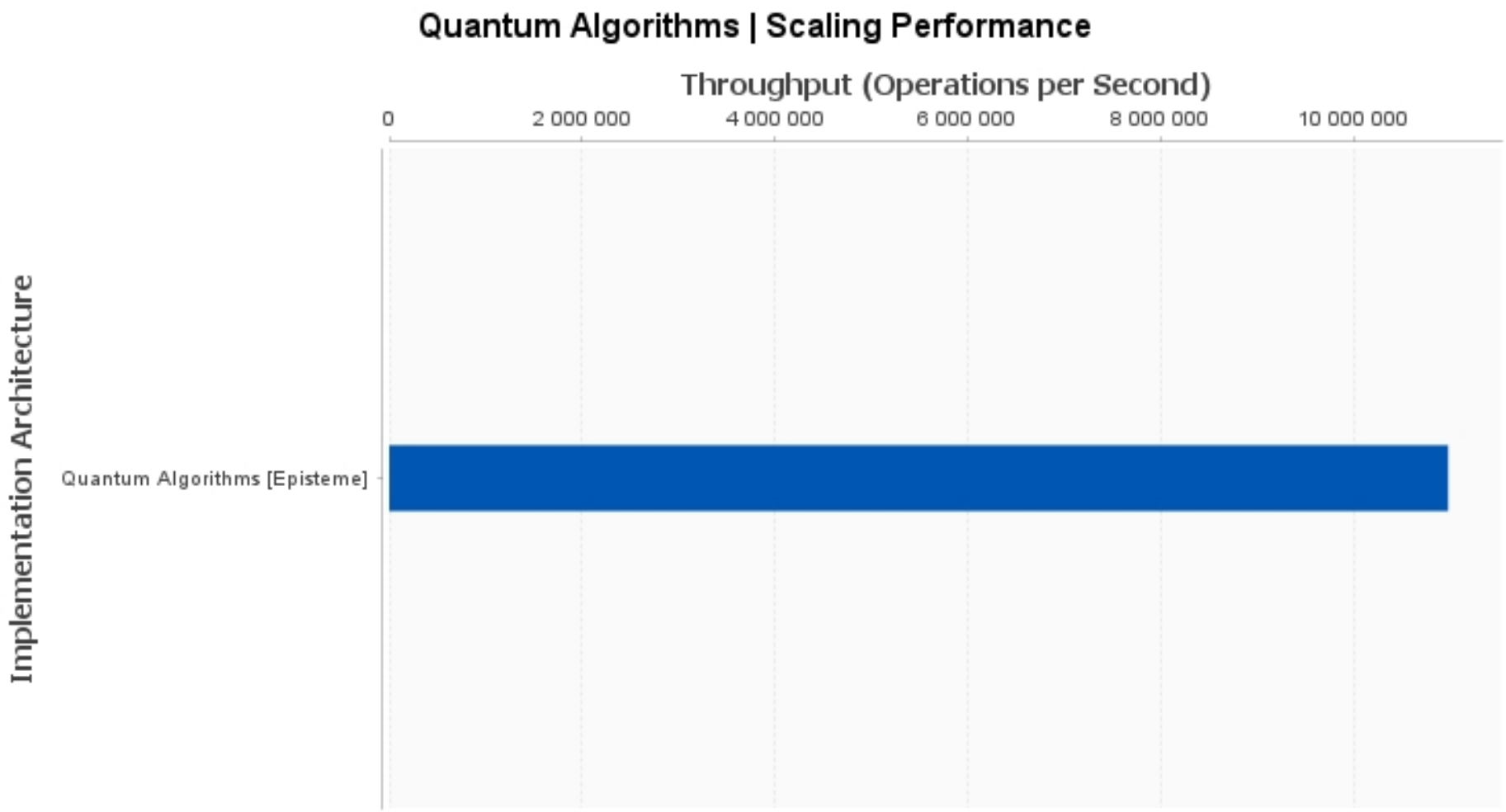
Domain Analysis: Parallel Computing



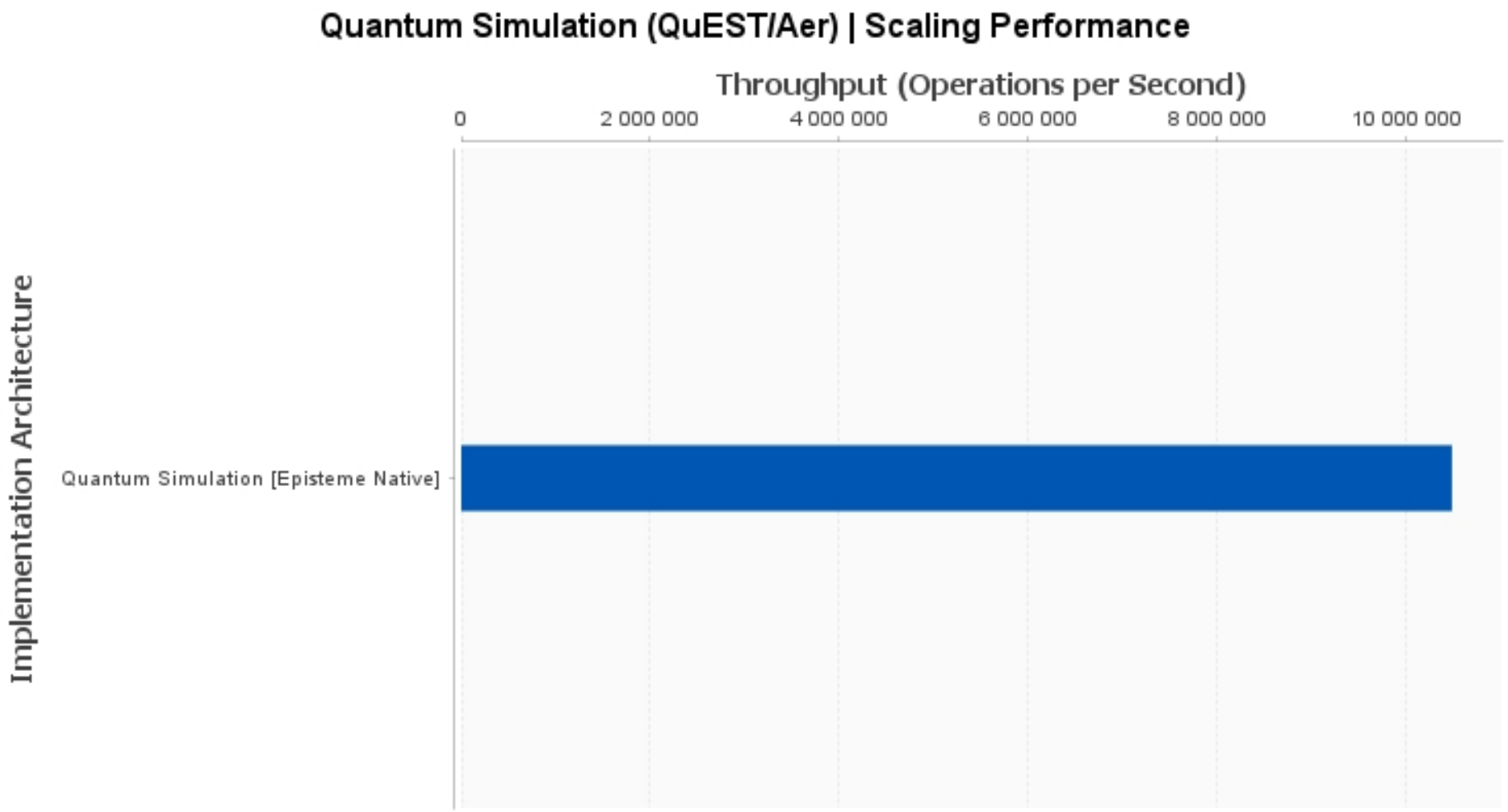
Domain Analysis: Quantum Algorithms



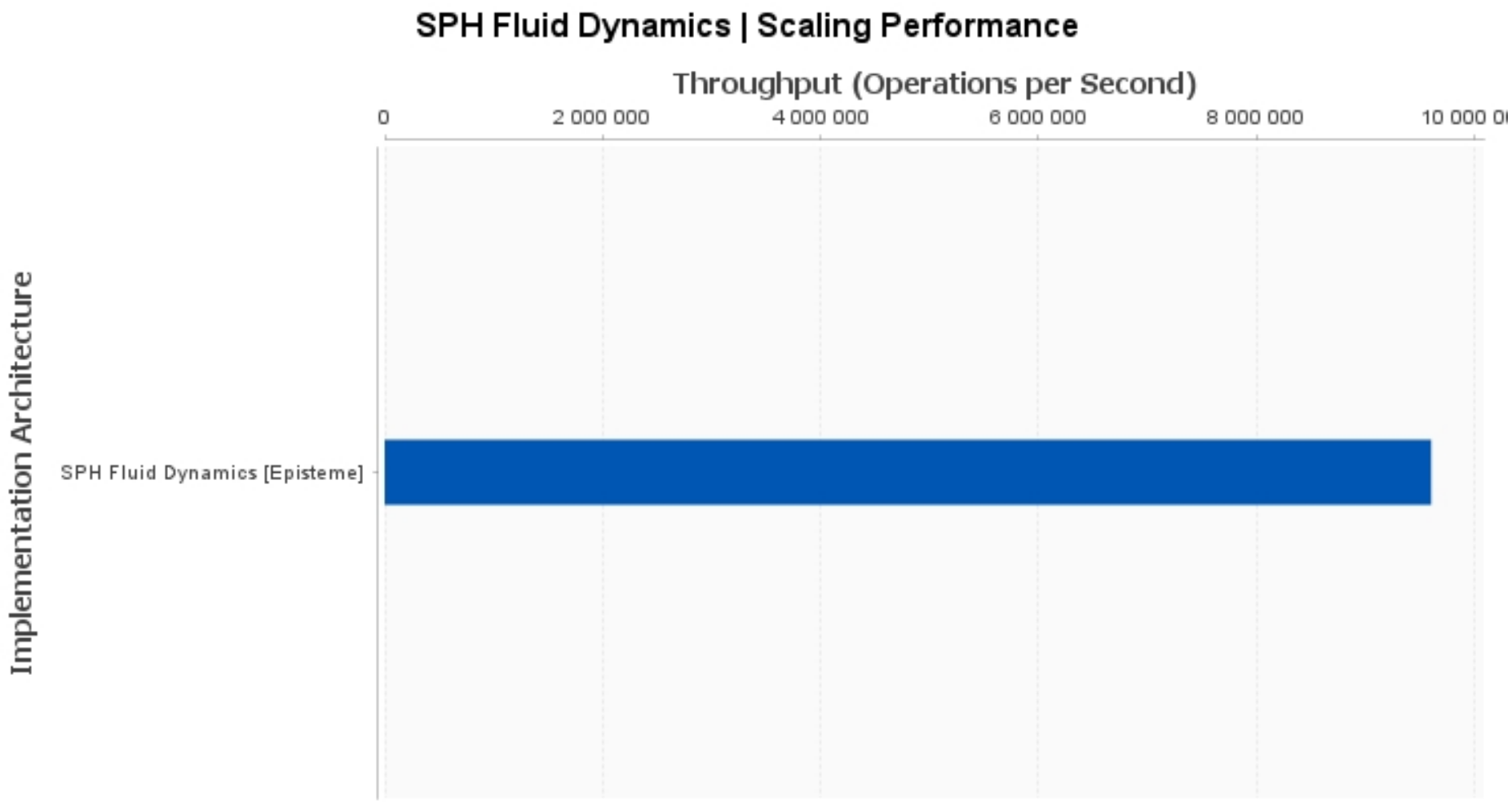
Domain Analysis: Quantum Simulation (QuEST/Aer)



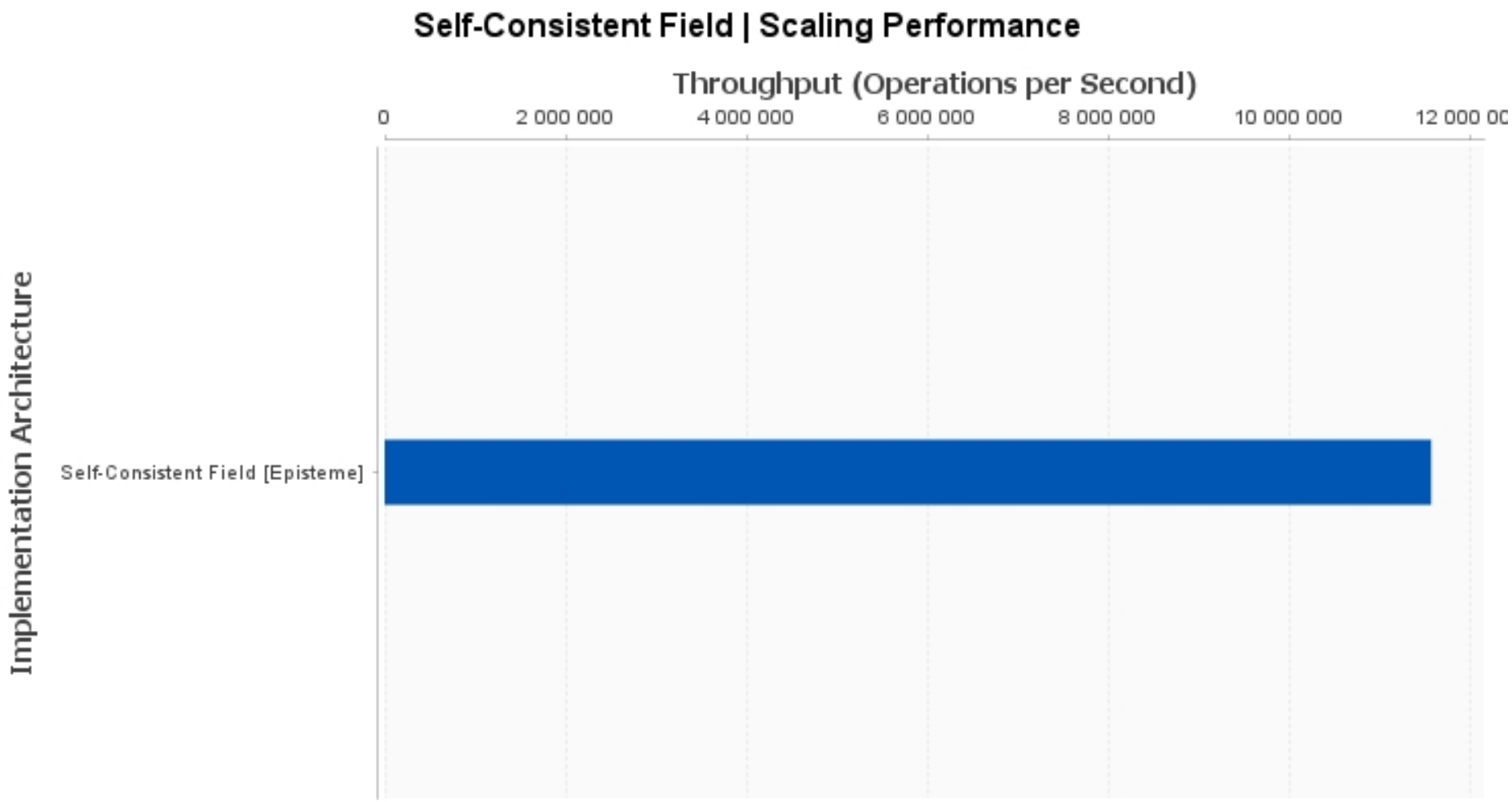
Domain Analysis: SPH Fluid Dynamics



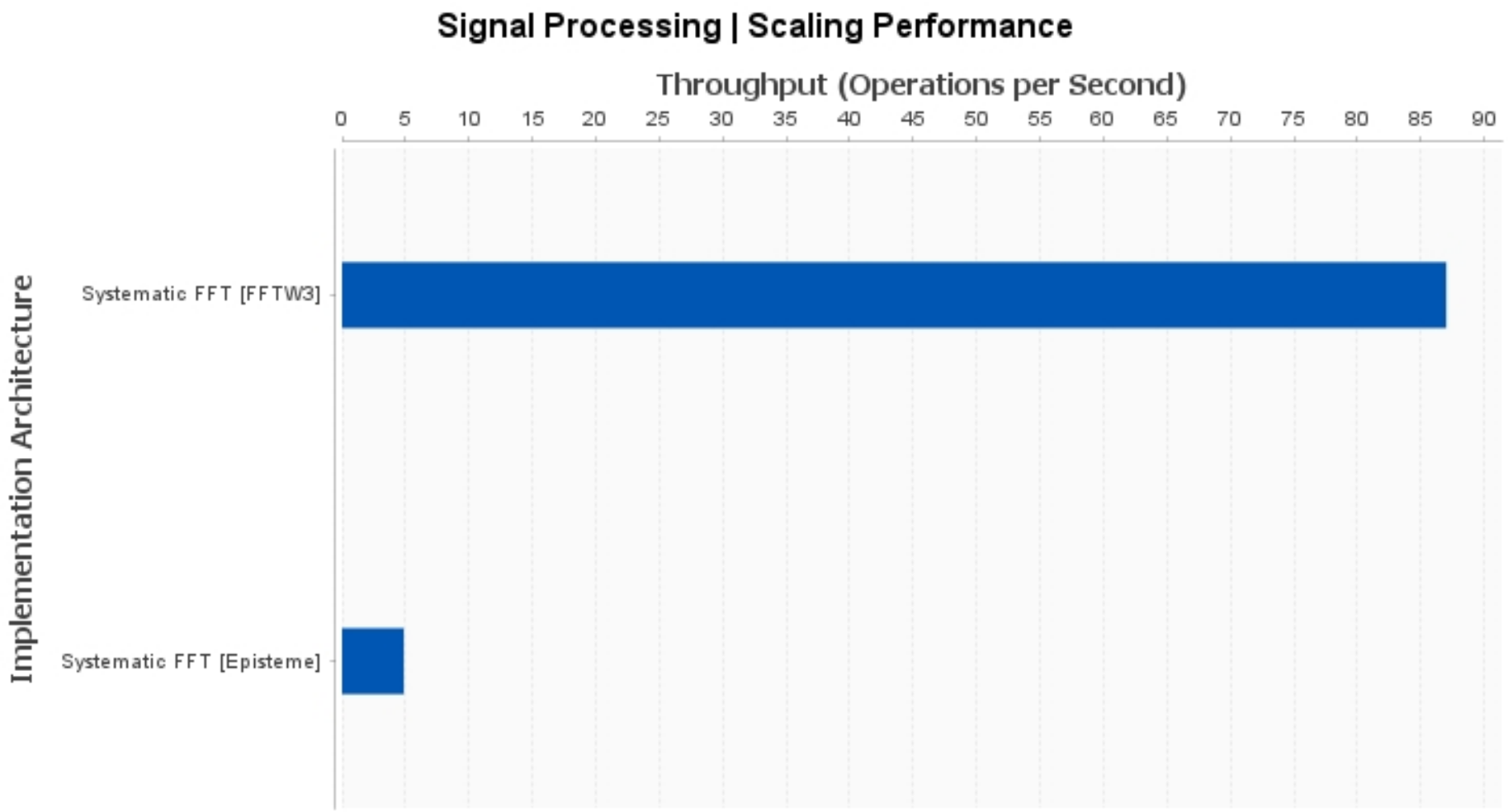
Domain Analysis: Self-Consistent Field



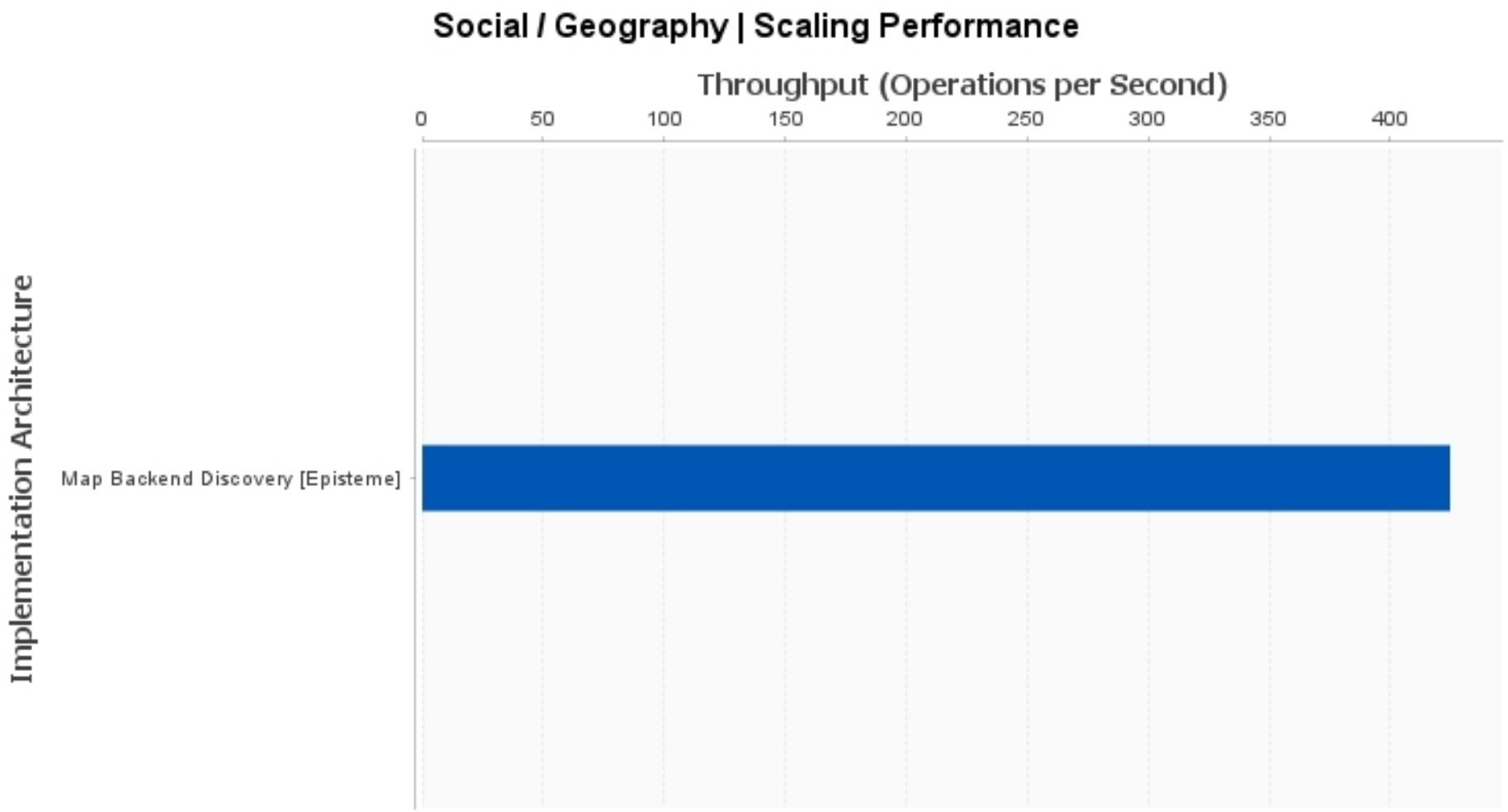
Domain Analysis: Signal Processing



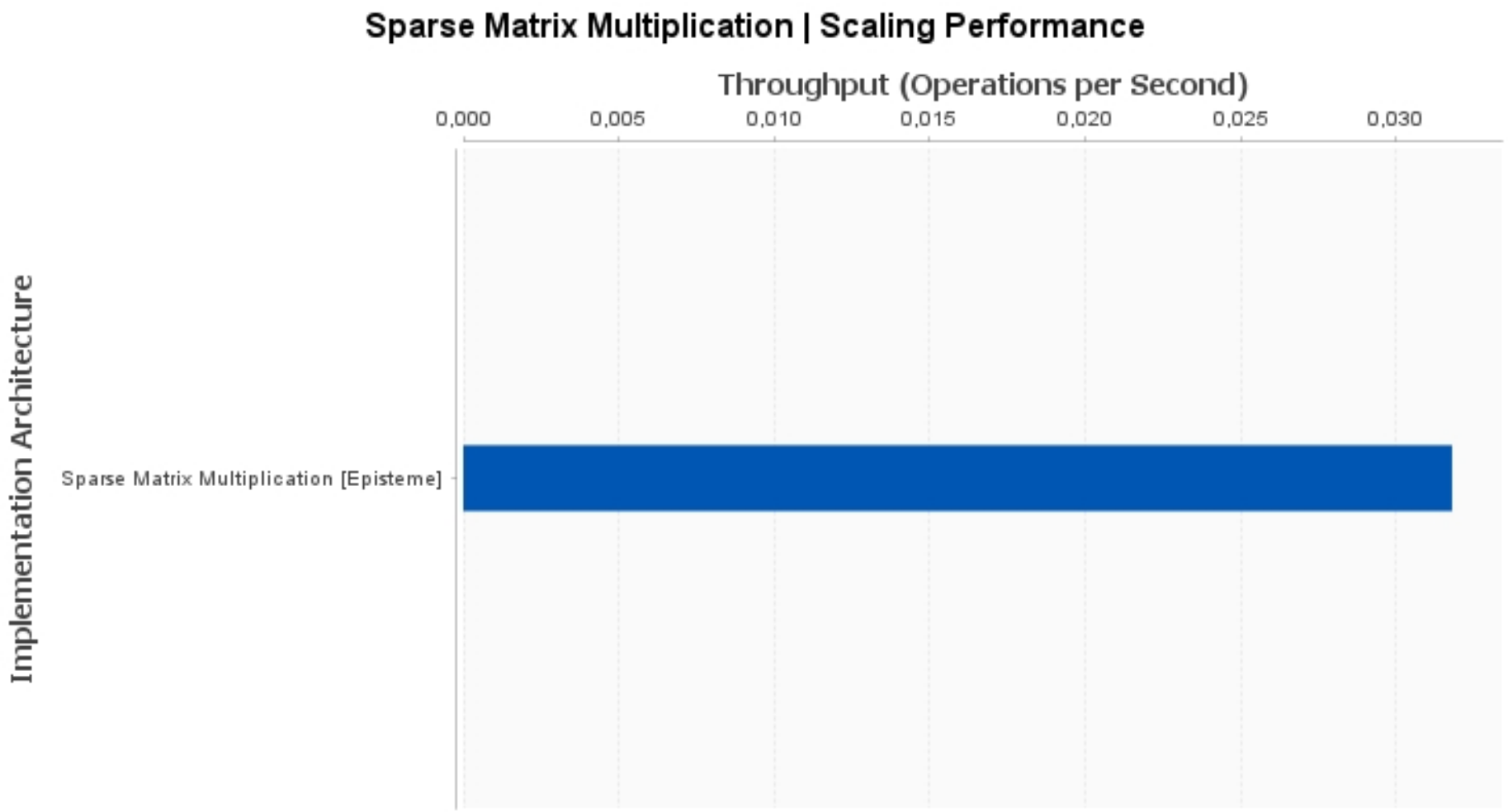
Domain Analysis: Social / Geography



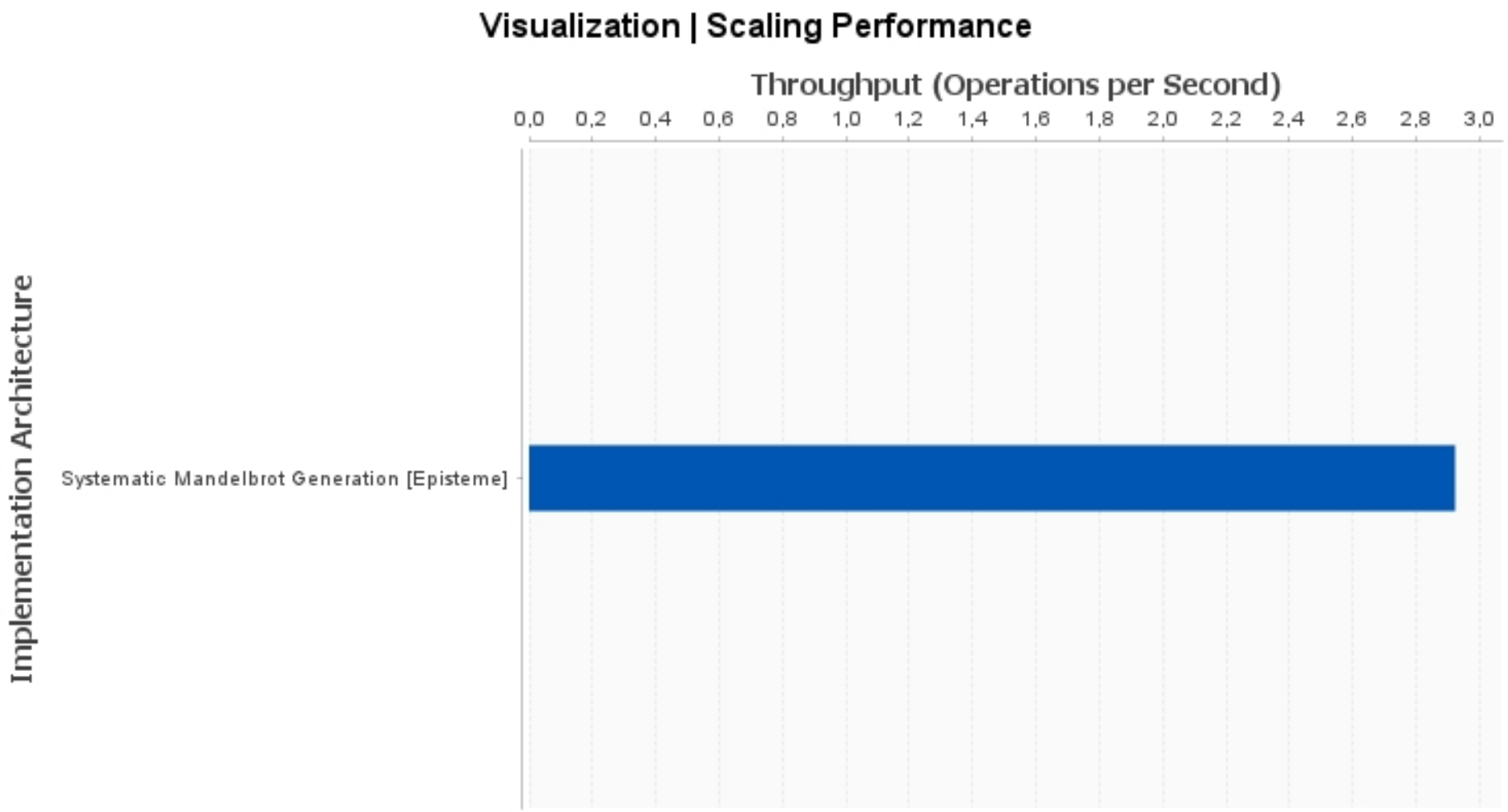
Domain Analysis: Sparse Matrix Multiplication



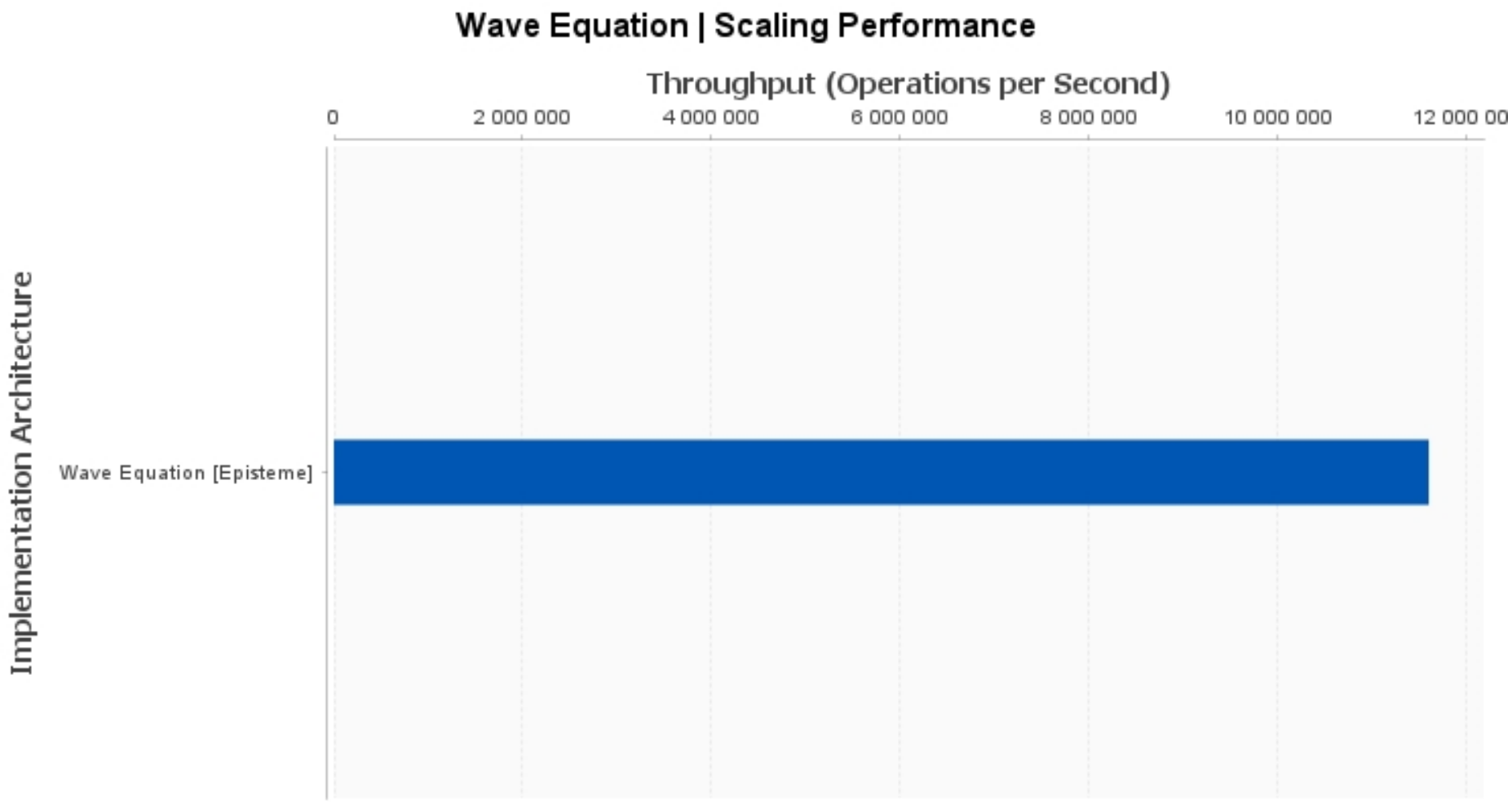
Domain Analysis: Visualization

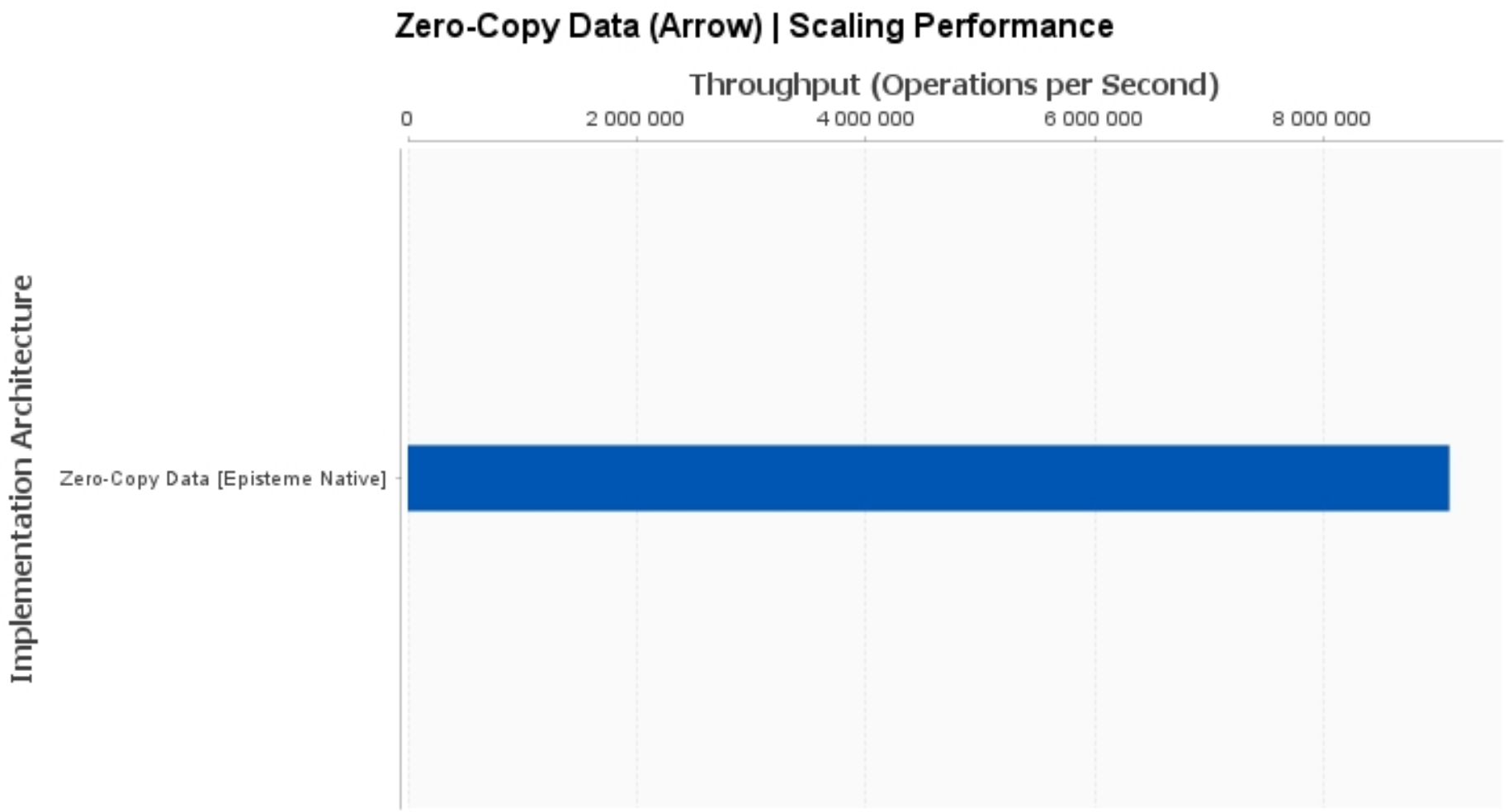


Domain Analysis: Wave Equation



Domain Analysis: Zero-Copy Data (Arrow)


Wave Equation | Scaling Performance
Throughput (Operations per Second)
0
2 000 000
4 000 000
6 000 000
8 000 000
10 000 000
12 000 00
Implementation Architecture
Wave Equation [Episteme]


Zero-Copy Data (Arrow) | Scaling Performance
Throughput (Operations per Second)
0
2 000 000
4 000 000
6 000 000
8 000 000
Implementation Architecture
Zero-Copy Data [Episteme Native]